

Classic and non-classic psychedelics for substance use disorder: A review of their historic, past and current research



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ABSTRACT

Substance use disorder (SUD) is a global public health concern that affects millions of people worldwide. Considering current research, addiction has been noted as the last stage of a chronic disease that may impair brain reward circuit responses and affects personal and social life. Treatments for SUD face challenges including availability and limited pharmacological response, often resulting in low retention of patients. A growing number of studies from the 'psychedelic renaissance' have highlighted the therapeutic potential of psychedelics for several psychiatric disorders, including SUD. In this non-systematic review we discuss past and current clinical and observational studies with classic (LSD, DMT, psilocybin and mescaline) and non-classic (ibogaine, ketamine, MDMA, salvinorin A and THC) psychedelics for the treatment of SUD published until December 2021. Although results are still inconclusive for LSD, DMT, mescaline, MDMA and Salvinorin A, in general, the literature presents moderate evidence on the controlled use of psilocybin and ketamine for Alcohol Use Disorder, ketamine for management of opiate and alcohol withdrawal, and THC preparations for reducing withdrawal symptoms in Cannabis and possibly in Opioid Use Disorder. Importantly, studies suggest that psychedelics should be more effective when employed as an adjunct therapy. Extensive research is warranted to further elucidate the role of psychedelics in the treatment of SUD.

1. Introduction

Harmful substance use represents a global public health concern that affects millions of people around the world. Health-related issues caused by drug misuse are symptoms of substance use disorder (SUD), which may include: compulsive and risky consumption of a substance; unsuccessful attempts to control consumption; cravings; persistent use despite the awareness of its harmful effects; impairment of social, professional, and recreational activities due to substance use; tolerance and withdrawal symptoms [1].

It is estimated that 36 million people suffered from SUD in 2019, with 500 thousand deaths attributed to drug use, including deaths due to liver cancer and cirrhosis, HIV infection and hepatitis C caused by unsafe injection practices, substance overdose and premature death [2].

Tobacco smoking was a leading factor for illness and death in 2015, causing over 7 million deaths [3]. It is estimated that 2.3 billion people aged 15 years and older worldwide are alcohol drinkers [4]. Mortality rates associated with harmful alcohol use are higher than those caused by diabetes, tuberculosis and HIV, with 3 million deaths in 2016 [4].

Addiction might be considered a chronic disease that may impair the brain's reward circuit capacity to properly respond to reward and the modulation of stress and self-regulation [5]. Potentially addictive substances increase dopamine (DA) release in the mesolimbic reward pathway either directly, acting within the system, or indirectly, by disrupting the modulatory activity of other neurotransmitters [6]. These substances stimulate dopaminergic neurons in the ventral tegmental area (VTA), which projects to nucleus accumbens (NAc), involved in appeti-

Abbreviations: 5-HT, 5-hydroxytryptamine; Δ^9 -THC, Δ^9 -tetrahydrocannabinol; AA, Alcoholics Anonymous; ASI, Addiction Severity Index; AUD, alcohol use disorder; BDNF, brain-derived neurotrophic factor; CBD, cannabidiol; CB, cannabinoid; CPP, conditioned place preference; CUD, cannabis use disorder; CREB, cAMP response element binding protein; DA, dopamine; DMT, N, N-dimethyltryptamine; ERK, extracellular signal regulated kinase; GDNF, glial cell line-derived neurotrophic factor; LSD, lysergic acid diethylamide; MAO, monoamine oxidase; MDA, 3,4-methylenedioxymphetamine; MDMA, 3,4-methylenedioxymphetamine; NAC, nucleus accumbens; NMDA, N-methyl-D-aspartate; OUD, Opioid Use Disorder; PFC, prefrontal cortex; RCT, Randomized Controlled Trial; RES, Randomized Experimental Study; SAU, smoking as usual; SUD, substance use disorders; THC, tetrahydrocannabinol; UDV, *União do Vegetal*; VTA, ventral tegmental area.

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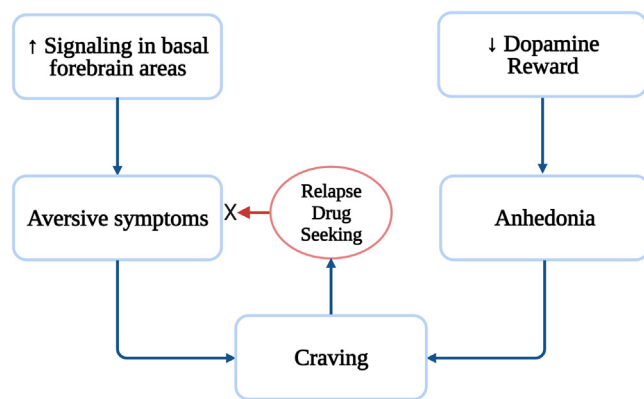


Fig. 1. The withdrawal phase. The signaling in basal forebrain areas (extended amygdala and habenula) is increased, triggering aversive symptoms. Non-drug related DA reward is decreased, causing anhedonia. These symptoms increase craving, which results in drug-seeking behaviors as a way to counteract the aversive state.

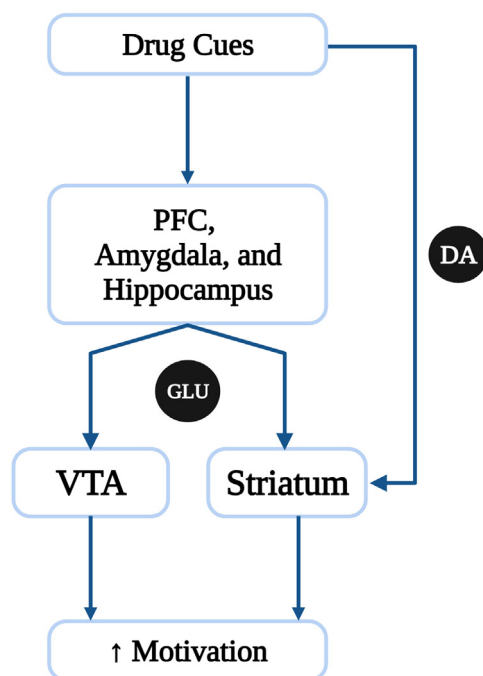


Fig. 2. General circuits underlying craving. Drug-seeking behavior is triggered by drug cues. DA in the striatum is increased and the glutamatergic projections from the PFC, amygdala and hippocampus to the VTA and striatum modulate motivation to use the substance. DA - dopamine; PFC - prefrontal cortex; VTA - ventral tegmental area; Glu - glutamate.

tive and aversive processes related to motivation [7]; the amygdala and hippocampus, related to emotional memory; and the prefrontal cortex (PFC), which mediates self-regulation [5]. Repetitive DA stimulation in these areas acts as positive reinforcement for drug intake [8] and creates a conditioned association of consumption and reward [9]. In addition, chronic use attenuates the drug-induced DA release and reduces the experience of reward [9], which can result in tolerance and the need of higher doses to achieve the same level of satisfaction.

Once substance effects wear off, the withdrawal phase begins (Fig. 1), in which intense distress, anhedonia, anxiety, and other physical symptoms may work as negative reinforcement and contribute to relapse, as the individual seeks the drug to avoid the unpleasant state [5,9]. Craving (Fig. 2) may happen independently or be present during the withdrawal phase. During cravings, drug cues are processed below

the threshold of conscious awareness by the neural systems that mediate drug memories [10], triggering conditioned responses. The orbitofrontal and anterior cingulate cortex are the circuits responsible for salience and value attribution, and the conditioned responses are mediated by the hippocampus and amygdala [9]. The sensitivity to drug cues and aversive states are modulated by glutamatergic projections from these structures to the VTA and striatum, eliciting the motivation to use the substance [9].

Continued drug consumption may induce the prefrontal cortical circuits underlying executive functions to be hypofunctional, whereas withdrawal and low sensitivity to nondrug rewards cause the limbic circuits underlying conditioned responses and stress reactivity to be hyperfunctional [9]. This combination affects motivational systems and impairs decision-making processes, resulting in distress and great difficulty for quitting substance use despite the negative consequences and decreased pleasure due to tolerance [9,11].

Although drug use has increased and presents a global public health concern, the availability of treatment is still limited and only one in eight people with SUD had access to treatment in 2019 [2]. The efficacy of treatments for SUD depends on a combination of medication and psychotherapy, given to complications related to mental health comorbidities and the difficulty of individuals to adhere to treatment programs [12,13]. The main pharmacological strategies in the treatment of SUD aims to reduce craving, decrease or prevent the positive reinforcement caused by the drug, and minimize or abolish the unpleasant symptoms caused by withdrawal (including craving, anxiety, depressive mood and other symptoms typical to each drug). In some cases, replacement therapy is also a useful strategy for harm reduction. The currently available medications include antagonists, partial or full agonists of receptors, and drugs with different pharmacological targets, such as enzymes that metabolize the substance consumed. Antagonists block receptors targeted by the substance and stop reinforcement of drug-affected neural pathways; partial and full agonists can mimic some of the effects produced by the drug and stabilize neural pathways that have been disrupted by harmful substance use [12,14]. Some examples of commonly used medications are: acamprosate and disulfiram to treat Alcohol Use Disorder (AUD); methadone and buprenorphine to treat Opioid Use Disorder (OUD); nicotine replacement therapy, bupropion, and varenicline to treat Tobacco Use Disorder; and naltrexone, used for both AUD and OUD [12,14]. Non-pharmacological interventions are also used and can be associated with pharmacotherapy. Psychotherapy involves different techniques that aim to modify behaviors that maintain substance use, such as Cognitive Behavioral Therapies [12], and can be offered as individual or group counseling, including mutual peer support programs, such as Alcoholics Anonymous (AA) [12]. The severity of the problem dictates whether outpatient or inpatient care should be offered [13].

Abuse and dependence of hallucinogen substances are still matter of debate. The use of natural drugs to alter the mind has been present in human history since remote times. There are reports of cannabis use and other hallucinogenic plants and mushrooms dated from thousands of years. Many of these substances were used as part of sacred rituals by Indigenous and traditional peoples [15–17]. In the last century, some synthetic hallucinogens were developed and became popular as recreational drugs, together with many natural hallucinogens. This class of substances can be described by many different terms: hallucinogens, mind altering drugs, psychedelics, psychodysleptics, psychotomimetics, entheogens, and entactogens, among others [17,18]. Despite some differences in the meaning of these terms, they are widely employed in a generic way. In this review, we will adopt the term psychedelics, which can be divided into two main groups: classic and non-classic.

Classic psychedelics are agonists or partial agonists of serotonin (5-hydroxytryptamine, 5-HT) 2_A receptors, their primary pharmacological target in the brain, as well as 5-HT $_{1A}$ and 5-HT $_{2C}$ receptors [19]. Their affinity to 5-HT receptors is related to their tryptamine moiety, which is the same scaffold in the chemical structure of serotonin [20].

The substances listed as classic psychedelics in this review are lysergic acid diethylamide (LSD), dimethyltryptamine (DMT), psilocybin and mescaline. The effects exerted by these psychedelics include profound subjective alterations in mood, thought and perception, visual and auditory hallucinations, synesthesia, mystical-type experiences, derealization, depersonalization, ego dissolution, etc. [21,22].

Non-classic psychedelics are substances with varied pharmacological mechanisms. For instance, ketamine is a N-methyl-D-aspartate (NMDA) receptor antagonist; 3,4-methylenedioxymethamphetamine (MDMA) acts as a 5-HT and DA reuptake inhibitor and releaser; ibogaine is an indole alkaloid with multiple targets; salvinorin A is a *kappa*-opioid receptor agonist; and tetrahydrocannabinol (THC) acts as a cannabinoid receptor partial agonist [23]. Although the classification of cannabis as a psychedelic substance is controversial, THC clearly promotes alterations on sensory perception and can induce mystical experiences, besides having important therapeutic properties [24,25]. Similarly to classic psychedelics, these substances can lead to altered states of consciousness with profound changes in somatic and affective processes, perception and cognition [23]. However, they have different primary mechanisms of action and their subjective effects profile is clearly distinguishable from those produced by classic psychedelics [22]. There are other nature-based hallucinogens, such as atropine and scopolamine which act as cholinergic antagonists or muscimol and ibotenic acid, gabaergic and glutamatergic agonists respectively, but these substances will not be addressed in this review.

The first boom in psychedelic research occurred in the years 1950–1970s. The Controlled Substances Act of 1970 classified psychedelics as Schedule I substances in the United States, considered dangerous due to their high potential for harmful use and no currently accepted medical use [20]. As a consequence, the National Institute of Mental Health stopped funding studies and the Drug Enforcement Agency hindered academics, leading to an abrupt interruption of research into the medical application of psychedelics [26,27]. During the same time, the International Psychotropic Convention of the United Nations of 1971 was being developed and similar restrictions were extended worldwide [26], which created political and cultural barriers against research involving psychedelics in many countries.

Despite the prohibitory status of psychedelics, an increasing interest in the therapeutic use of these compounds has been observed. Recent studies have shown the therapeutic potential of psychedelics for several clinical conditions, including SUD [23,28–36], in agreement with investigations conducted in the 1950s and 1960s. The therapeutic potential of psychedelics in the treatment of SUD is of interest, especially due to the involvement of other systems besides the dopaminergic system that might also suffer neuroadaptations that can lead to SUD, given their regulatory role in reward and reinforcement [5,6,9,11,37]. For instance, increasing evidence suggests that the opioid system may be associated to processes like mood modulation, drug consumption and reward circuits related to drug and food intake [38]. The glutamatergic, GABAergic, and endocannabinoid systems constitute other important pathways modulating affective, hedonic, and aversive circuits in the brain [39–41]. Therefore, psychedelics might represent a step forward in the development of new treatments since they act upon different neurotransmitter systems besides the dopaminergic system.

Treatment using psychedelics is usually referred to as Psychedelic-assisted Therapy, which consists of individual or in group continuous psychotherapy with a single dose of the psychedelic during a dedicated session in which the patient is encouraged to focus on how to overcome their SUD. Some studies also evaluated the effect of complementary doses. There are usually one or more follow-ups to check the long-lasting effects of the treatment, including interviews with family members about the patients' substance use status. The approach for the Psychedelic-assisted Therapy as reported in this review varies, especially in the early studies as there was not a well-established methodology for this type of trial [42]. In this review, we are going to present past and current data of classic and non-classic psychedelics, discuss their possi-

ble mechanisms of action, current state of evidence and limitations of their use in the management of SUD.

2. Methodology

We conducted a non-systematic review of clinical studies evaluating the therapeutic use of classic and non-classic psychedelics for SUD published until December 2021. First, we searched at the databases PubMed and Web of Science for the following terms: “drug addiction OR drug abuse OR substance use OR substance use disorder OR dependence” AND each of the following drugs: “LSD (OR lysergic acid diethylamide), DMT (OR dimethyltryptamine OR ayahuasca OR hoasca), psilocybin, mescaline, ibogaine (OR noribogaine), ketamine, MDMA (OR methylenedioxymethamphetamine), Salvinorin A (OR *Salvia divinorum* OR Salvia), and THC (OR tetrahydrocannabinol OR cannabis)”. Depending on the number of results, the search was limited selecting the filters to “clinical trial, randomized controlled trial, meta-analysis and systematic review” to remove non-human studies. The titles and abstracts were then screened to select the studies directly related to the topic of this review. The criteria of inclusion were papers written in English and with the full text available by the publisher or when it could be accessed by repositories or other sources. The exception was the search with ayahuasca that also included papers in Portuguese and Spanish because there are important studies in South American countries with this DMT-containing preparation. After reading the papers (original research and reviews), relevant citations were searched on the internet and included in this review, when their full content was available. This strategy was adopted because many past publications were not found in the PubMed/Web of Science search. The criteria of exclusion were papers written in languages different from those cited above, duplicated studies (only the original or the complete study was included) and when the full article could not be obtained.

2.1. Classification of studies

For each drug, clinical studies were classified according to the type of study and SUD evaluated. We attempted to classify the clinical studies as Randomized controlled trials (RCT), Randomized experimental studies (RES), Prospective pilot studies, Open label studies, Observational (longitudinal or cross-sectional) studies and Case reports, whenever possible; otherwise, we used the terminology adopted by authors to describe their experimental design. The main terminology for the reviewed clinical studies is available in a glossary at Box 1. We tried to get information regarding the experimental sample, presence or absence of control group, doses and posology used, and other aspects of the experimental design. When information was absent or inconclusive (dose, randomization, if inpatient/outpatient, adjunct interventions, etc.) it was not included in the tables. It was not our objective to perform any statistical analysis to validate the therapeutic use of the cited psychedelics, but we tried to discuss some degree of evidence for each substance on SUD, some positive and negative aspects and current limitations. Although not included in the result tables, online surveys were also discussed at the end of this review, because this kind of study has increased in the last years, despite some methodological issues that prevents more in-depth conclusions.

Preclinical studies were also screened, and the main studies were included in this review in order to bring additional evidence of the anti-addictive effect of psychedelics and to discuss possible mechanisms of action. However, such experimental studies were not classified or included in the tables.

3. Results and discussion

Fig. 3 shows the number of studies found for each psychedelic until December 2021 combining the substance names and the terms related to SUD (described in Section 2). The number of publications that match

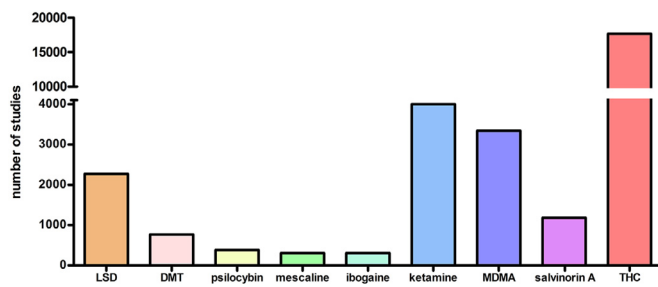


Fig. 3. Number of studies published up to 2021 for each psychedelic found in PubMed search combining the substance names and the terms “drug addiction OR drug abuse OR substance use OR substance use disorder OR dependence”, as described in Section 2.

our search criteria was higher for cannabis/THC, followed by ketamine, MDMA, and LSD. However, since the word cannabis was included in the search for THC, the results brought many studies not directly related with this review (for instance, studies with other cannabinoids). Also, most studies evaluated the prevalence of use, mechanisms of action or other pharmacological and social aspects of the substances, which is out of the scope of our review.

Only clinical, observational, and case reports studies that investigated the effectiveness of psychedelics for SUD were classified. Following our search and screening strategies we selected 76 clinical studies evaluating the use of psychedelics for SUD. Most psychedelics were evaluated for AUD. There were 10 RCT, 15 RES, 4 Prospective pilot studies, 13 Open label studies, 7 Observational (longitudinal or cross-sectional) studies (except anonymous online surveys) and 5 Case reports. Many clinical studies (most from 1950 to 1970s) had no clear experimental design and in some cases the methodology used was poorly explained in the text, which made it difficult to analyze these data. The following sessions bring more details about the studies performed with each psychedelic.

3.1. Classic psychedelics

3.1.1. LSD

Lysergic acid diethylamide was first synthesized in 1938 by Albert Hofmann as a potential circulatory and respiratory stimulant [43]. The commercialization of LSD by Sandoz laboratories under the trade name Delysid during the 1950s made it possible to use it in a variety of experiments [44]. In addition to its use in clinical research, LSD also became popular as a recreational drug during the 1960s as part of the counter-culture era, despite the negative impact on its reputation [27].

LSD is a mixed 5-HT₂/5-HT₁ receptor partial agonist, with affinity for various 5-HT receptors subfamilies - namely 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{1E}, 5-HT_{2A}, 5-HT_{2C}, 5-HT_{5A}, 5-HT₆ and 5-HT₇ [45] - and also DA receptors D₁ and D₂ [46,47]. It is a potent drug that can produce important alterations of consciousness at doses as low as 20 µg, with rapid absorption from the gut and primary metabolism by the liver [20].

The majority of studies investigating the role of LSD as a treatment for SUD focused on alcoholism. We were able to find 20 publications (Table 1), from which two are RCT, 6 are open label studies and 12 are not classified, most of them being reports of projects and treatments conducted in private and public clinics or hospitals. We also found several cross-sectional studies based on online surveys.

The history of LSD as a treatment for alcoholism goes back to 1953, when psychiatrists Humphry Osmond and Abram Hoffer introduced the drug to individuals with history of chronic alcohol use, testing the hypothesis that its capacity to help individuals achieve expanded levels of self-awareness could have therapeutic benefits [48]. In a small pilot experiment, two patients received a single dose of 200 µg LSD; one patient stopped drinking immediately after the treatment and remained sober

for 6 months, and the other quit drinking during the follow-up period [48].

Several clinical studies with LSD were then conducted on a substantial number of patients diagnosed with AUD. The first studies took place in the late 1950s [49,50] and were followed by the development of a treatment program based on the principles of AA [51]. The program consisted of weekly group therapy and AA meetings, work group therapy, movies with discussions, individual psychotherapy and the administration of a single dose of LSD [51]. Colin M. Smith [50] investigated the use of LSD and mescaline as adjuncts in the treatment of alcoholism in 24 patients receiving group therapy. There was no control group, and the follow-up period was one year, on average. Three of the 17 patients who received LSD (200–400 µg) achieved abstinence or very small rates of alcohol consumption, four patients decreased their alcohol intake, and 10 did not present significant changes. Smith [50] suggested that the drugs could be useful adjuncts to psychotherapy, as part of the treatment program, and speculated that the enhanced self-awareness appeared to have influenced the patients' behavior after the treatment.

A follow-up report by Chwelow et al. [49] described the treatment of 16 patients with an implemented version of the original program applied by Smith [50]. After an average follow-up of 6 months, 10 patients were abstinent, five reduced alcohol intake and one did not present significant changes. Authors suggested that surrender and self-acceptance were easily achieved in the LSD experience and associated such states to its therapeutic value. An open-label study with 61 patients who received a single dose of 400 to 1500 µg of LSD showed that 30 of them were abstinent and 16 of them presented reduction in alcohol consumption after the treatment [52]. In 1959, the Psychiatric Department of Moose Jaw Union Hospital, Saskatchewan, began to test LSD treatment for AUD patients [53]. O'Reilly & Reich [54] examined the effects of 200 µg on 33 participants of the program and reported that 17 of them benefited from the treatment, showing that LSD elicited profound alterations in perception that were therapeutically manageable. In a subsequent follow-up survey, O'Reilly & Funk [53] found that, from 68 patients treated between 1959 and 1962, 38% remained abstinent during the two months that preceded follow-up, but they did not find significant change in abstinence between the two follow-up periods.

A report of Psychedelic-assisted Therapy demonstrated the effect of LSD in patients with several psychiatric diagnoses, including AUD and SUD [55]. There was no control group and the dosage was gradually increased in each session, from 50 µg up to 150 µg. Patients presented some improvement after treatment, but statistical validation was not possible due to insufficient number of cases in the various subgroups. Besides that, the absence of a control or comparison group limits firm conclusions. Therefore, subsequent studies were more careful with methodological aspects of research and included comparison groups with no LSD.

Sven E. Jensen reported the results of a pilot project for treating patients with alcohol addiction with group therapy and a single dose of 200 µg of LSD during the last week of treatment, in comparison to group therapy or individual psychotherapy only [51]. Those treated with LSD were abstinent or showed significant improvement at the follow-up in comparison to the other two groups. A different report showed that 63% of the subjects treated with the same dose of LSD remained sober after discharge, compared to 27% in the control group, which only received individual treatment [56]. Kurland et al. [57] described the data of 69 pilot patients after the incorporation of a psychedelic session with 450 µg of LSD. One-third of patients remained abstinent for up six months and the authors concluded that LSD was relatively safe within the context of the treatment and the benefit in some cases seemed considerable.

Contrasting with these previous positive results, other studies did not show substantial improvement in AUD after LSD use, compared to controls. Cheek et al. [58] investigated the efficacy of two doses of LSD (100 µg followed by a second dose of 200 or 300 µg) plus group therapy versus a regular program. No statistically significant difference be-

Table 1
Human studies with classic psychedelics for substance use disorder.

LSD				
Substance	Type of study and sample	Intervention	Main results	Reference
Alcohol	Preliminary open label trial, <i>N</i> = 17, inpatient, 1-year follow-up.	Psychotherapy supplemented by occupational and recreational therapy + LSD (200–400 µg, orally). Other patients received mescaline (0.5 mg/kg).	Three individuals were completely abstinent, 4 reduced alcohol intake and 10 presented no change in drinking pattern.	Smith, 1958 [50]
Alcohol	Follow-up report, <i>N</i> = 33 (including 16 new patients and the sample of Smith, 1958), inpatient, 6–18-month follow-up.	Same protocol of Smith [50] (psychotherapy + LSD 200–400 µg orally), but with addition of auditory stimuli, visual stimuli, emotional stimuli and a change in the attitude of the people in contact with the patient.	From the new sample (<i>N</i> = 16), 10 were completely abstinent, 5 reduced alcohol intake and 1 presented no change in drinking pattern.	Chwelos et al., 1959 [49]
Alcohol	Open label trial, <i>N</i> = 61, inpatient, 18-month follow-up.	Psychotherapy + LSD (400 to 1500 µg, orally).	30 completely abstinent, 16 reduced alcohol intake, and 15 presented no change in drinking pattern.	McLean, 1961 [52]
Alcohol	Open label trial, <i>N</i> = 33, 88-week follow-up.	Psychotherapy + LSD 200 µg + additional 100 µg (if necessary).	7 completely abstinent, 10 reduced alcohol intake, 10 presented no change in drinking pattern and 6 were lost to follow-up.	O'Reilly & Reich, 1962 [54]
Alcohol	Report of pilot project for treating alcoholics, <i>N</i> = 138, 2 months (on average), inpatient, 18-month follow-up.	Group psychotherapy + LSD 200 µg compared to group psychotherapy without LSD and individual psychotherapy (control).	Half the patients of both groups not treated with LSD were lost to follow-up. Statistical analysis showed that patients in the LSD group were significantly more abstinent or improved at the time of follow-up compared to the other groups.	Jensen, 1962 [51]
Alcohol	Report of project for treating alcoholics, <i>N</i> = 115, 2 months, inpatient, 18-month follow-up.	Group therapy + LSD 200 µg compared to individual treatment by other psychiatrists (control).	Almost half the patients in the control group were lost to follow-up. Significantly more patients in the LSD group were abstinent or improved at the time of follow-up compared to those from the control group.	Jensen & Ramsay, 1963 [56]
Alcohol	Follow-up survey, <i>N</i> = 68, inpatient, 14-month follow-up.	Program of milieu therapy + LSD 200 µg.	38% abstinent in the two months preceding follow-up. No significant change in abstinence or indulgence.	O'Reilly & Funk, 1964 [53]
Alcohol	Open label controlled trial, <i>N</i> = 64, 4 weeks, inpatient, 12-month follow-up.	Group therapy + LSD orally (first dose of 100 µg and second dose of 200 or 300 µg one week apart) compared to regular program without LSD (control). The LSD session happened in groups of 8, where 2 took the drug and the other 6 gave support.	No statistically significant difference between groups at any follow-up period. Marked advantage for the LSD group at 3-month were lost by 6-month follow-up.	Cheek et al., 1966 [58]
Alcohol	Controlled study, <i>N</i> = 30, inpatient and day-care, 6-month follow-up.	LSD (800 µg) or ephedrine sulfate (60 mg – placebo group) + group psychotherapy. A third group (control) received the same procedures and therapies, but no drug session. Didactic meetings or group psychotherapy every morning and afternoon. Other services available: physio- and occupational therapy facilities, individual casework and psychiatric interviews.	The gain in abstinence time in the LSD, placebo and control groups was 34%, 32% and 20%, respectively. No statistically significant difference between treatment groups and no differential effect of LSD.	Smart et al., 1966 [61]
Alcohol	Report of an open-label study, <i>N</i> = 69, 3 weeks, inpatient, 6-month follow-up.	Psychotherapy + LSD 450 µg.	All changes on the clinical scales showed statistically significant beneficial impact (except for hypomania). One third of the patients maintained abstinence up to the six-month follow-up.	Kurland et al., 1967 [57]
Alcohol	Open label controlled trial, <i>N</i> = 99, inpatient, 18-month follow-up.	- Pilot study: adjustable dose of LSD (enough to evoke a reaction). - Control: patients who had been in the same program prior to the addition of LSD. - Experimental: 1, 2 or 3 sessions of LSD (100 to 800 µg - average dose of 400 µg).	No statistically significant difference between groups and no noticeable difference related to the number of sessions.	Van Dusen et al., 1967 [59]
Alcohol	Randomized controlled trial, <i>N</i> = 72, inpatient, 6-month follow-up.	LSD (600 µg, orally) or dextroamphetamine (60 mg, orally – control). No attempt at psychotherapy.	Both groups improved considerably. LSD patients were significantly better than the dextroamphetamine group at the 2-month follow-up, but no differences were observed at 6-month follow-up. Overall, no difference between treatments.	Hollister et al., 1969 [62]

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Table 1 (continued)

LSD				
Substance	Type of study and sample	Intervention	Main results	Reference
Alcohol	Single-blind study, <i>N</i> = 95, inpatient/outpatient, 1-year follow-up.	- LSD 1: 300 µg + adjustable dose 1 h later, intravenously. - LSD 2: 300 µg + adjustable dose 1 h later, intravenously + therapist present. - Comparison group: 3.75 g Sodium amobarbital + 30 mg of methamphetamine hydrochloride + adjustable dose 1 h later, intravenously + therapist present. - Control: Routine clinic care with therapy, without hospitalization. Split dosage technique: another injection 1 h later with a dose based on clinical assessment.	All groups improved significantly at 1-year follow-up, but there was no significant difference between groups.	Johnson, 1969 [63]
Alcohol	Controlled follow-up study, <i>N</i> = 176, 1 month, inpatient, 12-month follow-up.	Comparison of different interventions: - Hypnodelic: milieu therapy + LSD 3 µg/kg orally + hypnosis + psychotherapy during drug session. - Psychedelic: milieu therapy + LSD 3 µg/kg orally + psychotherapy during drug session. - LSD alone: milieu therapy + LSD 3 µg/kg orally, without any other intervention. - Control: milieu therapy only. Half of the patients in each group received disulfiram just prior to and after discharge, for the period of 1 year.	Significant improvement on most measures for all groups, but no significant difference between groups. 50–60% of patients had relapsed at 3-month follow-up. Disulfiram made no difference in terms of treatment outcome.	Ludwig et al., 1969 [66]
Alcohol (and Neurosis)	Open label controlled trial, <i>N</i> = 24, inpatient and outpatient, 12-month follow-up.	Repeated doses of LSD, ranging from 50 to 300 µg, orally, until the effect was judged adequate. Control group had no intervention. Patients received a 5 mg tablet of dextroamphetamine before the drug session to reduce anxiety.	The supposed therapeutic benefits of LSD treatment could not be demonstrated by statistical analysis of the results. The analysis of the results was hampered because many patients did not complete the 6-month and 12-month follow-up.	Denson & Sydiaha, 1970 [60]
Alcohol	Follow-up study, <i>N</i> = 140, 2 months, inpatient, 1-year follow-up.	Experiment I: - LSD: 500 µg orally + human relations training. - Control: human relations training only. Experiment II: - LSD: 500 µg orally + human relations training. - Placebo: 25 µg orally + human relations training. - Control: human relations training only. The human relations training was based on exercises to enhance interpersonal problem-solving.	Experiment I: no statistical difference between groups, although both showed good adjustment 1 year after treatment. Experiment II: ratings of good adjustment higher, but no differential effect of treatments on outcome. Analysis of both experiments: no statistical difference between LSD + human relations training and human relations training alone.	Bowen et al., 1970 [64]
Alcohol	Controlled study, <i>N</i> = 333 (113 schizophrenics, 220 nonschizophrenics), 90 days, 12-month follow-up.	LSD (500 µg) + regular program compared to regular program alone (control 1). A second control group had taken the regular program previously. The protocol of regular program was not described.	The benefits of all treatments disappeared in the course of a year. Nonschizophrenic LSD-treated patients had the highest abstinent percentages. Marked negative effect of LSD upon schizophrenic patients regarding post discharge drinking control. Nonschizophrenic patients who volunteered for LSD but did not receive it had the worst results.	Tomsovic & Edwards, 1970 [67]
Alcohol	Randomized controlled trial, <i>N</i> = 135, 6 to 7 weeks, 18-month follow-up.	Psychotherapy + LSD high-dose (450 µg) or low-dose (50 µg – placebo group). The program offered up to 3 sessions, but the majority of patients had only 1 session.	Psychedelic psychotherapy successfully helped more than half of the alcoholics treated. Significant small advantage of high-dose group at 6 months disappeared at 12 and 18 months.	Kurland et al., 1971 [65]

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Table 1 (continued)

LSD				
Substance	Type of study and sample	Intervention	Main results	Reference
Alcohol and narcotics	Report of Psychedelic-Assisted Therapy, <i>N</i> = 17.	Psychedelic-assisted Therapy with an average of 6.2 sessions per patient with an interval of 2 weeks on average. Gradual increase of LSD dosage: - 50 µg in the 1st session. - 25 µg more in the 2nd session. - 150 µg total by the 4th or 5th session. - Drop to 100 µg after patient has learned to work with the drug.	The addicted patients presented some improvement in symptoms, productivity and anxiety.	Chandler & Hartman, 1960 [55]
Heroin	Controlled study, <i>N</i> = 74, 4 to 6 weeks, inpatient and outpatient, 12-month follow-up.	Regular clinic program (control) or psychotherapy + LSD (300 to 450 µg) + regular clinic program. Psychedelic therapy was inpatient, whereas the regular clinic program was outpatient and included group therapy. Regular clinic program was based upon the “abstinence approach,” with weekly group therapy + daily urine surveillance.	The mean follow-up ratings showed no statistically significant differences between groups. High correlation of abstinence with adjustment.	Savage, 1973 [68]
DMT / Ayahuasca				
Substance	Type of study and sample	Intervention / survey methodology	Results	Reference
Crack-Cocaine	Cross-sectional observational study, <i>N</i> = 40 crack-cocaine users and former users.	Qualitative semi-structured interviews using the DSM-IV criteria for dependence. Ayahuasca doses were not clearly reported.	80% of participants ceased drug consumption after starting using ayahuasca. The others 20% reported having fallen back into crack/cocaine use after having attained abstinence with the help of ayahuasca. Among the reasons for relapse, detachment from the ayahuasca community stood out.	Cruz & Nappo 2018 [90]
Multiple substances	Cross-sectional observational study, <i>N</i> = 84.	Interviews with adolescents regarding the use of a variety of psychoactive substances, comparing ayahuasca users and non-users (controls). Ayahuasca doses were not clearly reported.	Ayahuasca users were mostly comparable to control except for a considerably smaller consumption of alcohol in the past and at the time of the study. Religious affiliation may have played a central role as a possible protective factor for alcohol use.	Doering-Silveira et al., 2005 [84]
Multiple substances	Longitudinal observational study, <i>N</i> = 242, 12-month follow-up.	Addiction severity was assessed using the Addiction Severity Index (ASI), and records of history of the alcohol and illicit drug use. Two subsamples: - Study 1: jungle-based sample; - Study 2: urban-based sample. Ayahuasca doses were not clearly reported.	Ayahuasca users showed lower scores on the ASI - Alcohol Use and Psychiatric Status subscales. Jungle-based ayahuasca users showed a higher frequency of previous illicit drug use but ceased at the time of examination, except for cannabis. 12-month follow-up showed abstinence from illicit drug use was maintained in both groups except for cannabis in the jungle-based sample. Differences on ASI scores were still significant in the jungle-based group but not in the urban group.	Fábregas et al., 2010 [83]
Multiple substances	Longitudinal observational study, <i>N</i> = 12, inpatient, 6-month follow-up.	Four days of group counselling combined with two expert-led ayahuasca ceremonies. Data was collected by several psychological and behavioral factors related to problematic substance use and qualitative information from their experience. Ayahuasca doses were not clearly reported.	The treatment provided significant improvements in scales assessing hopefulness, empowerment, mindfulness and quality of life meaning and outlook subscales. Self-reported alcohol, tobacco and cocaine use declined, although cannabis and opiate use did not. Reported reductions in problematic cocaine use were significant. All study participants reported positive and lasting changes from participating in the retreats.	Thomas et al., 2013 [85]
Multiple substances	Longitudinal observational study, <i>N</i> = 53, inpatient.	Integrative program with sessions of detoxification, diet and ayahuasca ritualistic sessions, together with sessions of psychotherapy. Data was collected by several scales, like, ASI and Craving Experience Questionnaire (CEQ). Ayahuasca doses were not clearly reported.	A significant decrease in addiction severity outcomes, drug use, alcohol use, psychiatric status, emotional distress and substance craving was observed. Quality of life increased significantly from beginning to treatment completion. First indications for significantly improved SUD symptoms after the Amazonian medicine-based treatment.	Berlowitz, 2019 [92]

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Table 1 (Continued)

DMT / Ayahuasca				
Substance	Type of study and sample	Intervention / survey methodology	Results	Reference
Multiple substances	Longitudinal observational study, <i>N</i> = 11, inpatient, 6-month follow-up.	Four days of retreats with two expert-led ayahuasca ceremonies. Data obtained by semi-structured interviews. Ayahuasca doses were not clearly reported.	All participants reported reductions in substance use and craving; 8 participants reported complete cessation of at least one substance at follow up. Increased connectedness with self, others and nature/spirit was described as a key element associated with reduced substance use and craving.	Argento et al., 2019 [86]
Multiple substances	Longitudinal observational study, <i>N</i> = 36, inpatient, 2 to 12-month follow-up.	Integrative program with Amazonian traditional treatments and Western psychotherapeutic methods. Data was collected by several scales, like ASI and CEQ. Ayahuasca doses were not clearly reported.	Clinically positive changes were found across all repeated measures. The changes appeared early in the treatment and were maintained over time. Significant improvements were also found for neuropsychological functioning.	O'Shaughnessy et al., 2021 [91]
Psilocybin				
Substance	Type of study and sample	Intervention	Results	Reference
Alcohol	Open-label study, <i>N</i> = 10, within subject, no control group, 12-week outpatient treatment; 36-week follow-up.	12-week course psychosocial treatment with psilocybin oral administration at week 4 (0.3 mg/kg) and week 8 (0.3 or 0.4 mg/kg).	Adverse effects were mild and transient. Percent drinking days and heavy drinking days decreased during weeks 5–12 relative to baseline. Alcohol abstinence increased significantly after psilocybin administration and was maintained at 36-week follow-up.	Bogenschultz et al., 2015 [112] (Qualitative analysis in Nielson et al., 2018 [113])
Alcohol	Case report (Preliminary descriptive case study of an ongoing double-blind placebo controlled clinical trial).	Protocol for psilocybin group was similar to that described in Bogenschultz et al. [112], but with dose of 25 mg/70Kg (week 4) and 30 or 40 mg/70 kg (week 8). Control group received diphenhydramine (50 mg, week 4; 100 mg, week 8; orally). The dose of week 8 could be kept the same of week 4 both for psilocybin and diphenhydramine, depending on the response on the first session. An additional open-label session with psilocybin (25–40 mg/70 kg) was offered to all participants who meet interim safety after the double-blind period (34 weeks after randomization).	Individual experiences and alterations in the self-perception, consciousness and in the relationship with alcohol and drinking were reported as possible reasons for alcohol abstinence by three patients from the ongoing study (the participants' group remained blinded).	Bogenschultz et al., 2018 [114]
Nicotine (tobacco)	Open-label study, <i>N</i> = 15, no control group, 15-week outpatient treatment.	4 weeks of cognitive-behavioral therapy, with the first dose of psilocybin on week 5 (20 mg/70 kg), second dose on week 7 (30 mg/70 kg) and an optional third dose on week 13 (30 mg/70 kg) during a structured 15-week smoking cessation treatment protocol.	No clinically significant adverse events. At 6-month follow up 80% of the volunteers showed seven-day point abstinence evaluated by self-report and biomarkers.	Johnson et al., 2014 [109] (Qualitative follow-up in Noorani et al., 2018 [111])
Nicotine (tobacco)	Follow-up of the previous study (Johnson et al. [109]).	As described in Johnson et al. [109].	67% of participants were smoking abstinent at 1-year follow-up and 60% remained abstinent on a long-term follow-up (> 16 months).	Johnson et al., 2017 [110]
Mescaline				
Substance	Type of study and sample	Intervention	Results	Reference
alcohol	Pilot study, <i>N</i> = 24, no control group, inpatient, 2-month to 3-year follow-up.	Single oral dose of mescaline (0.5 mg/kg, <i>N</i> = 7) as adjuncts to superficial psychotherapy supplemented by occupational and recreational therapy. Other participants received LSD (200–400 ug, <i>N</i> = 17).	From 7 participants treated with mescaline, two were classified as unchanged in alcohol intake, two improved and three much improved.	Smith, 1958 [50]
alcohol	Case report of a member of the Native American Church.	Not applicable.	According to the case report, the individual stopped drinking alcohol after attending the peyote meeting.	Albaugh et al., 1974 [123]
Heroin and dilaudid	Case report, <i>N</i> = 3.	Not applicable.	Describes that individuals were able to quit drug use after the peyote ritual.	Blum et al., 1977 [126]

tween groups was found during 12 months of follow-up, although LSD patients showed advantage over controls at 3 months, but it was lost by 6 months. Another trial by Dusen et al. [59] compared different groups: one from the pilot phase in which subjects received an adjustable dose of LSD (ranging from 100 to 800 µg); three experimental groups who received 1, 2 or 3 sessions of LSD; and a control group who had been in the same program prior to the addition of LSD. No significant difference among groups and no noticeable difference related with the number of sessions were observed. Denson & Sydiaha [60] conducted an open-label trial with people with AUD and other patients with psychiatric disorders who were treated with repeated doses from 50 to 300 µg. Results and discussion did not disclose outcomes by their disorder, therefore it is not possible to know the exact outcome for the AUD patients. Additionally, because almost half the patients were lost to follow-up, possible benefits of the LSD treatment could not be fully assessed.

Some researchers also included a comparison group treated with a non-psychedelic substance besides a control group who did not receive any drugs. In a single-blind controlled study patients were divided in three groups, one that received 800 µg of LSD, one 60 mg of ephedrine sulfate, and one that received therapy but no drugs [61]. After a six-month follow-up abstinence time had increased among all groups with no significant difference between them, suggesting that LSD was no more effective than other treatments. A RCT compared a single dose of 600 µg LSD with a single dose of 60 mg dextroamphetamine and a no-drug control group [62]. The results suggested considerable improvement for both groups with an advantage for LSD patients at two-month follow-up but discontinued at six months. The authors stated that there was no substantial difference among treatments and LSD did not seem suitable as a primary treatment, although it might help patients through alleviation of guilt and “letting go”. Johnson [63] compared the effect of LSD (300 µg) with 3.75 g of sodium amobarbital (sodium amytal) plus 30 mg of methamphetamine hydrochloride (Methedrine) or routine clinic care (control group) in a single-blind trial. All groups improved significantly at one-year follow-up, but there was no statistical difference among groups. The author concluded that LSD in this setting was no better than more conventional therapy and speculated that the improvement caused by both drugs might be due to some neurophysiological mechanism as a result from strong affective discharge.

It is important to note that it is notably difficult to work with control and placebo groups when testing psychedelics, given that such substances have intense and distinguishable effects. A strategy researchers adopted was to administer a very small dose of LSD as “placebo” (also called active placebo) - enough to elicit a reaction but not enough for participants to experience the intense effects of the substance. A follow-up study by Bowen et al. [64] tested the efficacy of LSD for AUD in two different experiments. Experiment I was single-blind and compared participants that received 500 µg of LSD and a human relations training program, with participants that only received the training program, whereas experiment II was double-blind and included a third group in which patients received active placebo (25 µg of LSD) and the training program. No differences were observed between groups. Kurland et al. [65] also compared an active placebo dose of 50 µg against 450 µg of LSD in which patients received up to 3 sessions of psychedelic therapy. There was immediate improvement in both groups, but no significant differences between them. Researchers suggested that psychedelic psychotherapy was helpful to more than half of the treated participants, although a no-LSD control group prevented the differentiation of psychotherapy and LSD session.

Other researchers tried a unique approach using a specialized therapeutic technique called “hypnodelic therapy” combined with LSD (3 µg/kg). They tested four different treatments: hypnosis + LSD + psychotherapy during the drug session; LSD + psychotherapy; LSD alone; and milieu therapy only (see details in Ludwig et al. [66]). The results showed significant improvement on most measures for all groups both at inpatient and follow-up evaluations, but no significant difference between groups. At the three-month follow-up 50–60% of patients had

relapsed. Authors suggested that the LSD treatment did not produce any greater therapeutic benefits in comparison with the other treatments tested. Tomsovic & Edwards [67] evaluated the effect of a single overwhelming experience with 500 µg of LSD as a treatment for AUD in schizophrenic patients compared to controls who did not take the psychedelic. Despite some initial improvement, all groups showed loss of the progress made over a year and authors suggested that LSD had a negative effect on schizophrenic participants with AUD regarding post discharge drinking control.

We also found one paper investigating the efficacy of LSD as a treatment to heroin addiction [68]. In this controlled study, patients received a single dose of 300 to 450 µg and therapy, and the control group received group therapy only. The mean follow-up ratings showed improvement, but no statistically significant differences between groups, suggesting that LSD therapy might contribute to the rehabilitation of heroin addicts, but it was not possible to attribute the positive effects to the drug alone.

Despite numerous studies investigating the benefits of LSD for patients with SUD, we only found a few recent studies evaluating the anti-addictive capacity of LSD in animal models. Alper et al. [69] evaluated the effect of a single dose of LSD (25 or 50 µg/kg) on the self-administration of alcohol in C57BL/6 J mice using a two-bottle choice drinking paradigm. Ethanol consumption and preference decreased relative to controls for 50 µg/kg LSD, with a group mean reduction of 17.9% in ethanol consumption, which was sustained across 46 days. However, total fluid intake between control and LSD showed no statistically significant difference. Authors concluded that the efficacy of LSD involves psychological factors and biologically mediated effects [69]. Another preclinical study investigated the effects of 0.08 or 0.32 mg/kg LSD in Wistar rats using an alcohol deprivation effect model with prophylactic treatment scheme, but the results showed no significant difference between the experimental groups [70].

It is important to note that we could not find recent controlled trials investigating the therapeutic benefits of LSD for SUD, probably due to the legal and bureaucratic limitations around psychedelic research. Nevertheless, even the clinical studies prior to the Controlled Substances Act were not able to achieve consensus in the scientific community. While some studies showed promising results [48,50–52,54,56,57], others were not able to confirm the efficacy of LSD nor show significant differences between this psychedelic and other treatments [53,55,58–68]. Moreover, it is not yet clear whether LSD is effective as a treatment by itself or if it enhances the efficacy of more conventional treatments when used as an adjunct.

3.1.2. DMT

N, N-dimethyltryptamine is an indole alkaloid widely found in nature, both in plants and endogenously in some animals [71]. DMT is a potent 5-HT_{2A} agonist, besides having affinity for 5-HT_{1A}, 1B, 1D, 5-HT_{2C} and 5-HT₇ receptors [19,72–74], the latest involved in the modulation of synaptogenesis and neuritogenesis. Some studies have also observed an affinity of DMT with sigma-1 receptors, which may relate to a possible neuroprotective effect [75,76]. DMT is orally inactive because it is metabolized by the enzyme monoamine oxidase (MAO), present in the liver and intestine, as well as in the human brain. Thus, the psychedelic effect of this substance is observed only when it is ingested parenterally as a snuff or enema (although synthetic DMT can also be injected or smoked) [77].

Studies involving DMT are limited because the substance remains in the Schedule 1 of the Controlled Substances Act and due to the absence of effects when it is used orally in its pure form. Although there is a lack of data about pure DMT as a therapeutic alternative, ayahuasca, a drink containing DMT, has been receiving increasing attention as a potential therapeutic tool and will be the focus of this session.

Ayahuasca (from Quechua “Vine of Souls”) is the common name used in the literature to characterize a psychedelic botanical drink used for rituals and therapeutic purposes by different Amazonian tra-

ditional peoples and modern syncretic doctrines [71]. The beverage (a brew) is commonly prepared by boiling the barks and stems of the vine *Banisteriopsis caapi* together with the leaves of the shrub *Psychotria viridis*, although other plants can be used [16,18,78]. The *B. caapi* vine has β -carboline alkaloids including harmine, tetrahydroharmine, and harmaline; while the leaves of *P. viridis* contain DMT [79]. The β -carbolines inhibit MAO type A in the gastrointestinal tract and CNS, causing a potent synergistic effect, allowing DMT to access the circulatory system and cross the blood-brain barrier inducing the psychoactive effect.

The therapeutic potential of ayahuasca, despite being historically observed by traditional communities, only appeared in scientific literature in the past decades. Studies suggest that ayahuasca has therapeutic potential for anxiety disorders, depression and SUD [35,76]. Although there are some controlled clinical trials with ayahuasca for depression and anxiety, most studies related to SUD are observational, both qualitative and quantitative (Table 1).

In an observational report, Grob et al. [73] evaluated a group of 15 ayahuasca users in which 11 reported having a history of moderate to severe use of alcohol and other drugs before joining “União do Vegetal” (UDV), a Brazilian syncretic religion. According to this report, all users stopped using the drugs without reports of relapse, shortly after the start of recurrent use of ayahuasca. Employing ethnographic interviews and participant observation methods, Labigalini [80] qualitatively examined the subjective experience of four former alcohol users from Grob’s study and concluded that alcohol use was not replaced by ayahuasca, but instead, the individuals could expand their conscience and change their attitudes. Barbosa et al. [81] conducted a study with more than 1900 members of UDV who had problems associated with alcohol and tobacco. The authors showed that the success of the therapy among participants in ayahuasca rituals was superior to that found in other religions that work with people with SUD but did not have ayahuasca as part of the intervention. The study also showed that disorders related to alcohol and tobacco consumption were significantly lower in UDV members when compared to the average of the Brazilian population [81]. Perkins et al. [82] also observed that the use of ayahuasca in naturalistic settings seems to be related to lower self-reported current consumption of alcohol and other drugs for those with and without prior SUD.

Another study assessed SUD symptoms and related problems among ayahuasca users utilizing the Addiction Severity Index (ASI) and interviews [83]. In the first part of this study, the authors compared a group of ayahuasca users living in forest areas to controls living in rural areas; then, they compared urban-based ayahuasca users to urban controls. The authors conducted the follow-up after one-year and in both samples ayahuasca users scored significantly lower than their controls in the ASI and Psychiatric Status subscales, and these results were maintained in the follow-up tests [83].

A study conducted with adolescents belonging to UDV showed less use of alcohol and other drugs than young people of the same age who never used ayahuasca [84]. An observational study carried out in Canada evaluated the effect of participating in two sessions of assisted therapy with ayahuasca in combination with four sessions in counseling groups on harmful substance use. The researchers reported decreased use of alcohol, tobacco, and cocaine by the participants, but not for marijuana and morphine [85]. There was also an improvement in the mood and quality of life scales, suggesting that the therapy produced positive psychological and behavioral changes on participants. A qualitative study with 11 adults taking part in an ayahuasca retreat as a treatment for SUD showed that the intervention helped them to identify patterns of negative thought and individual barriers related to their SUD. All participants reported decreased use and craving, along with 72% of them ceasing at least one substance in a subsequent follow-up [86]. Mercante [87] conducted a study in three Brazilian communities that use ayahuasca for SUD. The researcher suggests that treatment with ayahuasca differs from substitution therapy because context situations are addressed, such as social issues, and the treatment provides a social

and personal transformation to the patients, decreasing their chances of relapse.

Other qualitative and observational studies also suggest the therapeutic potential of ayahuasca to treat different types of SUDs [88,89], including crack/cocaine addiction [90]. The Center for Rehabilitation of Drug Addicts and Research on Traditional Medicines – Takiwasi – localized into the Peruvian Amazon region, has played an important role in the evaluation of ayahuasca and other traditional medicines. An inpatient observational study conducted with 36 participants showed significant and clinically positive changes in their perceived stress, craving frequency, mental illness symptoms, spiritual well-being, and physical and emotional health. These changes appeared early in the treatment and maintained over time [91]. Another study conducted at the Takiwasi center assessed the efficacy of a short-term therapeutic program in a sample of 53 individuals diagnosed with SUD (most presented polysubstance use of cannabis, alcohol, and cocaine-related drugs). After the ayahuasca session, participants showed significant decrease in SUD symptoms and improvement of quality of life compared with baseline data [92]. The therapeutic potential of ayahuasca in SUD treatment was also assessed in a qualitative study conducted with health professionals and indigenous healers who use the beverage professionally in their interventions in the treatment of SUD. The study showed that, in carefully structured environments, ayahuasca may improve neurobiological and psychological processes that support the recovery from SUD and the prevention of relapses [93].

We did not find controlled clinical trials evaluating the effect of ayahuasca on SUD. On the other hand, some preclinical research seems to ratify the therapeutic potential of ayahuasca registered in observational studies and help to better understand the mechanisms of how ayahuasca can assist SUD therapy. Oliveira-Lima et al. [94] investigated the acute and chronic effect of ayahuasca (50–500 mg/kg) on behavioral sensitization induced by ethanol in mice submitted to the open field apparatus. Their findings suggest that all tested doses of ayahuasca decreased the animals’ acute and sensitized behavior. Another study using the same behavioral test showed that ayahuasca (corresponding to 1.76 mg/kg of DMT) attenuated the ethanol-induced behavioral sensitization in mice. In addition, researchers observed that ayahuasca reduced the anxiogenic response to ethanol withdrawal and prevented some ethanol-induced changes on 5-HT_{1A} density and prodynorphin levels in the hippocampus [95]. Cata-Preta et al. [96] investigated the potential of ayahuasca (30–300 mg/kg) and its isolated plants - *B. caapi* (132–1320 mg/kg orally) and *P. viridis* (3.75–37.5 mg/kg intraperitoneally) in the experimental model of conditioned place preference (CPP) in mice. The research identified that only ayahuasca blocked the development of ethanol-induced CPP and not the plants *per se*. Another study showed that ayahuasca (500 mg/kg orally) induced CPP *per se*, but its pre-treatment blocked the ethanol-induced CPP [97]. Nolli et al. [98] investigated the effect of ayahuasca (171–686 mg/kg) on voluntary ethanol intake of rats followed by a subsequent analysis of c-Fos immunoreactivity in areas related to SUD. Ayahuasca didn’t change the voluntary ethanol intake in the animals tested, however, patterns of c-Fos immunoreactivity changed with the lowest dose of ayahuasca.

Interaction of ayahuasca with other potentially addictive substances was also evaluated in preclinical studies. Godinho et al. [99] evaluated the effect of ayahuasca (2 ml/kg) on voluntary amphetamine intake, in addition to protocols related to behavior associated with anxiety and locomotor activity in adolescent rats. Authors suggested that ayahuasca modifies voluntary amphetamine consumption and prevents the effects related to anxiety and locomotor activity. A recent study showed that ayahuasca (100 mg/kg) and methylphenidate induced CPP *per se*, and ayahuasca blocked methylphenidate-induced CPP reinstatement [100]. Additionally, methylphenidate induced differences in FOS immunoreactivity in several brain regions, but ayahuasca had limited effects. Another study evaluated the ability of ayahuasca administration to prevent cocaine-induced behavioral sensitization [101]. Ayahuasca containing 1.76 and 3 mg/kg of DMT did not reverse behavioral sensitization. How-

ever, in a subsequent protocol, higher doses (15, 30 and 45 mg/kg of DMT) prevented the induction of behavioral sensitization by cocaine [101].

Preclinical and observational studies reporting the use of ayahuasca to treat SUD have encouraged the scientific community to perform more robust and controlled studies; however, it is important to emphasize that ayahuasca is a product prepared by the decoction of at least two plants with different chemical compounds and it is not clear whether the pharmacological therapeutic is due to the synergistic action of the plants or if their active principles also have a role themselves. An observational study investigated the role of *B. caapi* vine alone (without the presence of DMT) in the therapeutic context of 390 patients at the Takiwasi center [102]. Results suggest that the vine also has therapeutic potential by itself possibly due to the psychoactive action of β -carbolines, which might contribute to patients beginning to be aware of the reasons behind the drug use behavior. This study is important to be mentioned as it supports that *B. caapi* may enhance the therapeutic properties of ayahuasca, and ratifies the therapeutic and psychoactive effects of MAO inhibitors *per se*.

Along with the progress in psychedelic neurobiological science, data from ayahuasca studies indicate a promising aid in the treatment of depression, anxiety, and SUD. In this way, a group of researchers are conducting a study to create a method to evaluate and systematize research and treatment with ayahuasca. The Ayahuasca Treatment Outcome Project highlights the importance of treatment setting in the design, delivery, and efficacy of SUD rehabilitation programs with ayahuasca. The project employs qualitative research methods to determine the most important aspects of the settings to patients, practitioners, and, primarily, how they correlate with some outcome measures [103]. Initiatives like this, added to results of current research, encourage further studies to better understand the therapeutic potentials of these substances.

3.1.3. Psilocybin

Psilocybin (4-phosphoryloxy-N,N-dimethyltryptamine) is the main psychedelic compound found in *Psilocybe mexicana* and many other species of mushrooms known as magic mushrooms, sacred mushrooms and *teonanacatl* (Aztec name meaning god's flesh) due to theirentheogenic effects. The ritualistic, religious and social use of magic mushrooms by primitive American people (especially by the Mayans and Aztecs) has been well documented [18]. In addition to their spiritual and ritualistic uses, *Psilocybe* mushrooms have been used for healing purposes by primitive and Indigenous people and more recently by the general population both for recreational and medical purposes.

Psilocybin is an indole alkaloid that after absorption is dephosphorylated into its active metabolite psilocin, an indolamine whose structure is very similar to serotonin, what explains its effects as a potent serotonergic agonist [104]. Psilocybin and psilocin are agonists of serotonin binding to different subtypes of 5-HT receptors, mainly 5-HT_{2A}, with moderate effect on 5-HT_{1A} and 5-HT_{2C} [23,105,106].

Most studies with psilocybin date back to the decades of 1950 to 1970, but a substantial interest in this substance has emerged in the last 10 years [23,105]. The literature suggests that psilocybin is relatively safe if used in moderate doses and controlled environments. According to a review published by Tyliš et al. [104], psilocybin accentuates the pre-existing mood at very low doses, induces a well-controlled altered state of consciousness at moderate doses, while high doses may evoke strong psychedelic experiences. The substance remains in the Schedule 1 of the Controlled Substances Act, classified as dangerous and without medical use by the United Nations Convention on Psychotropic Substances [104]. On the other hand, recent studies have suggested the potential of psilocybin for several mental health conditions, including depressed mood, anxiety disorders (as well as anxiety and depression associated with terminal / life-threatening diseases), obsessive-compulsive disorder, and SUD [34,105,107]. Clinical studies evaluating the use of psilocybin for SUD are shown on Table 1. We found two open-label stud-

ies and one case report describing preliminary results of an ongoing clinical study.

The potential effects of psilocybin to treat SUD have been recently evaluated for alcohol and tobacco [108]. Johnson et al. [109] conducted an open-label pilot study to determine the safety and feasibility of psilocybin as an adjunct to tobacco smoking cessation treatment. Fifteen smokers attended four weekly meetings consisting of Cognitive Behavioral Therapy for smoking cessation and preparation for psilocybin administration before the administration of the first moderate dose of psilocybin (20 mg/70 kg). The meetings continued weekly, and the participants received another dose of psilocybin (30 mg/70 kg) at week 7, and an optionally extra dose at week 13. The authors reported that at the 6-month follow-up 80% of the participants showed seven-day point abstinence evaluated by self-report and biomarkers [109]. A further evaluation showed that 60% of the volunteers remained abstinent at 2.5-year follow-up [110]. Participants of this study were invited to a retrospective follow-up interview about 30 months after initial psilocybin sessions in order to identify perceived mechanisms of change leading to smoking cessation and key themes in participant experiences and long-term outcomes to better understand the therapeutic process [111]. Participants reported having vivid insights into self-identity and reasons for stop smoking from their psilocybin sessions and reported a range of persisting positive changes beyond smoking cessation, including increased altruism and pro-social behavior.

Bogenschutz et al. [112] conducted an open-label proof of concept study to evaluate the use of psilocybin for AUD. Ten participants received a 12-week psychosocial intervention that included psilocybin administration at weeks 4 and 8. The psychosocial interventions consisted of 7 sessions of Motivational Enhancement Therapy, three preparation sessions and two debriefing sessions supporting the psilocybin sessions. Several questionnaires were used to measure mood, motivation, craving, abstinence and scales of drinking behavior. Percent drinking days and heavy drinking days decreased after the psilocybin administration relative to baseline and weeks 1–4 (psychosocial treatment prior to psilocybin first session). Abstinence increased significantly following psilocybin administration with data suggesting decreases in craving and increases in abstinence self-efficacy, and these improvements were maintained at the 36-week follow-up [112]. Key phenomena related to change in drinking outcomes and acute subjective effects of the psilocybin session are presented in a qualitative analysis of debriefing sessions from this study [113]. A double-blind placebo-controlled clinical trial of psilocybin-assisted treatment for AUD is being conducted by the same group. So far, the authors have reported the qualitative descriptive case studies of three participants of this ongoing trial to illustrate the range of experiences and persisting effects of psilocybin treatment [114]. Results of this clinical study is supposed to be published soon and will further contribute to the applications of psilocybin in the treatment of AUD.

A preclinical study using Wistar rats evaluated three schemes of intraperitoneal psilocybin treatment on alcohol relapse-drinking, employing the long-term voluntary alcohol consumption with repeated deprivation phases [70]. In the first experiment rats received five injections of 1 mg/kg starting before the reintroduction of the alcohol bottles, followed by four applications in 12 h intervals; in the second experiment two high doses (2.5 or 10 mg/kg) were administered twice during the deprivation period; in the third experiment they evaluated the effect of eight chronic intermittent microdosing administration (0.1 mg/kg) during abstinence on subsequent relapse-like drinking behavior. Psilocybin (1 mg/kg – first schedule) acutely reduced the ethanol intake on the first day post-abstinence (alcohol re-presentation) but had no long-lasting effects on ethanol intake compared to the control group indicating no anti-relapse effect [70]. The other two schemes of treatment had no effect on ethanol intake after the abstinence time. These experiments did not reproduce the positive anti-alcohol effect observed in human studies, but we should note that it cannot satisfy the criteria for a translational model as it is not possible to offer psychosocial intervention on animal

models. Furthermore, the study employed a different route for psilocybin injection (intraperitoneal instead of oral).

Some neurochemical mechanisms to explain the therapeutic effects of psilocybin and other psychedelics have been proposed, while many researchers suggest that psilocybin induces experiences with personal meaning and spiritual significance with sustained positive changes in attitudes and behavior [115]. These experiences are persistent and strong enough to stimulate motivation to overcome the addiction. Recent studies using magnetic resonance imaging and positron emission tomography have investigated the ability of psilocybin to modulate functional brain connectivity and suggest that psilocybin induces time-dependent changes with long-lasting effects in cerebral functions [116–118]. How these changes can contribute to the therapeutic effects of psilocybin should be considered in future research.

3.1.4. Mescaline

Mescaline (3,4,5-trimethoxyphenylethylamine) is a phenylethylamine found in the Peyote cactus (*Lophophora williamsii*) and other species of cacti. The peyote is also known as divine cactus, divine herb, flesh of God, medicine of God, among other names and grows in desert regions of the southwestern United States and northern Mexico [119]. Other mescaline containing plants include *Lophophora diffusa* (false peyote, *peyote querétano*), *Pelcyphora aselliformis* (peyotillo or little peyote) from Central America, *Trichocereus pachanoi* (San Pedro cactus, traditionally known as *Wachuma*), and *Trichocereus peruvianus* (Peruvian Torch), which grow in Andean countries of South America [119]. Mescaline is an agonist of 5-HT₂ receptors, especially the 5-HT_{2C} subtype, but also acting on 5-HT_{2A} and 5-HT_{2B} subtypes with lesser potency [120,121]. Other secondary targets may be involved, including noradrenergic and dopaminergic systems (see Cassels & Sáez-Briones [120] for further details). The drug has low lipophilicity and crosses the blood–brain barrier with greater difficulty; therefore higher doses are needed to produce effects similar to other psychedelics [121,122]. However, effects last longer compared to psilocybin and LSD [119].

Peyote has been used by native peoples of Mexico for thousands of years due to its psychedelic properties. It is also used by several native American groups from the Native American Church, Pentecostal Christianity and Traditional Navajo Healing groups in religious and healing ceremonies, as the ingestion of peyote facilitates communication with God or divine entities due to its power to make the individual transcend the boundaries of earth, space and time [119]. With the renaissance of the psychedelics, peyote and mescaline have become popular again as a psychedelic substance by non-indigenous and non-religious people.

Members of the Native American Church report that mescaline is used to relieve symptoms of alcohol withdrawal and reduce relapse [36]. It is important to highlight that the use of peyote by native American people was generally confined to the rituals, even for people under some kind of treatment. Recent studies have proposed the use of mescaline for the treatment of depression, anxiety, obsessive compulsive disorder and drug addiction [35,121]. However, we only were able to find a few old reports supporting the use of mescaline for SUD (Table 1). The use of mescaline to treat AUD is mainly based on reports which show that members of the Native American Church and other traditional groups use peyote in their ceremonies as part of the treatment of members with history of alcohol abuse [123–125]. It is described that peyote induces a state of self-understanding through ritualized introspection and self-examination that leads to a turning point, in which the individuals have visions of the potential alcohol problems and their life without drugs [124,126].

There are a few case reports about the use of peyote for SUD among traditional people. Albaugh & Anderson [123] briefly reported a case of a member of the Native American Church that stopped drinking alcohol after attending the peyote meeting, but no additional details were provided. Blum et al. [126] described three cases of SUDs (two of heroin and one with dilaudid, an opiod used to treat moderate to severe pain) in which the individuals could quit substance use after the peyote rit-

ual, according to the authors. Despite the reported ritual use of peyote to treat AUD and other SUDs, there are no recent clinical trials or other scientific studies validating the use of mescaline for SUD without the religious setting. Instead, many authors believe that mescaline is effective only under the religious context, which is supported by anthropological studies [124].

We found only one clinical study evaluating the effect of mescaline in AUD conducted as part of a psychedelic treatment for alcoholic patients in the 1950s, but the methods and results were superficially described. A group of particularly refractory alcoholics were selected to receive LSD or mescaline as adjuncts to treatment consisting of superficial psychotherapy supplemented by occupational and recreational therapy [50]. Seven patients received a single dose of 0.5 g of mescaline orally and were evaluated on follow-ups conducted after two months to three years. Two patients showed no significant differences, two demonstrated substantial reduction in alcohol intake and three achieved complete abstinence. The author emphasized that this was a pilot study, and the use of mescaline or LSD was only part of the treatment program. A recent online survey evaluated the effect of mescaline used in naturalistic setting on psychiatric conditions and SUD [127]. The majority of respondents with a prior psychiatric condition reported improvements in these conditions following mescaline (76% for AUD and 68% for SUD). We did not find other studies that evaluated the use of mescaline for AUD or other SUDs.

3.2. Non-classic psychedelics

3.2.1. Ibogaine

Ibogaine (10-metoxibogamine) is the major indole alkaloid from *Tabernanthe iboga*, known for its psychoactive effects and first associated with anti-fatigue, cardiac regulatory, aphrodisiac, and neurasthenic properties [16,128]. Ibogaine was first isolated in 1900 by Édouard Landrin [129], and in 1970 it was classified as an FDA Schedule I drug, which resulted in a lack of investigation until the 1990s, when ibogaine had its use approved for clinical trial protocols [130]. Concurrently, Howard Lotsof reported improvements in withdrawal and craving symptoms in a group of heroin users that received 6–19 mg/kg of ibogaine [130,131]. Despite therapeutic findings, ibogaine remains illegal in the US mainly due to its side effects, which include neurotoxicity and QT-interval prolongation during electrocardiogram monitoring [128,132]. Recently, the therapeutic effects of ibogaine have been studied for the treatment of opioid, nicotine and alcohol use disorder, presenting promising results in both animal models and clinical studies [128,131,133]. However, we could only find one randomized placebo-controlled pilot study and five reports or retrospective studies collecting data of patients who received ibogaine in detoxification clinics in an open label basis (Table 2).

Ibogaine has a polypharmacological profile acting as a serotonin and dopamine reuptake inhibitor, 5-HT_{2A} agonist, NMDA non-competitive antagonist, nicotinic acetylcholine non-competitive antagonist and opioid agonist; presenting affinity for κ -, μ - and σ -opioid receptors [128,130]. Its metabolism occurs primarily through CYP_{2D6} liver enzymes that convert this alkaloid into the active metabolite noribogaine (12-hydroxyibogaine), associated with the long-lasting psychological effects usually related to ibogaine administration [128,132].

He et al. [134] proposed that the mechanism underlying the anti-addictive effects of ibogaine involves the increase in glial cell line-derived neurotrophic factor (GDNF) in the VTA. Systemic administration of ibogaine at a dose of 40 mg/kg reduced ethanol ingestion both in two-bottle preference test and operant conditioning chamber tests, demonstrating its potential in continuous access and auto administration contexts, respectively. The same study also observed a long-lasting reduction of ethanol intake after intra-VTA injection of ibogaine [134]. These findings are consistent with another study in which ibogaine administration at a dose of 40 mg/kg increased the expression of brain-derived neurotrophic factor (BDNF) and GDNF transcripts in dopamin-

Table 2
Human studies with non-classic psychedelics for substance use disorder.

Ibogaine				
Substance	Type of study and sample	Intervention	Main results	Reference
Opioids - heroin	Series of open label cases studies, $N = 33$, 48–72 h duration, outpatient.	Ibogaine 19.3 ± 6.9 mg/kg (6–29 mg/kg)	76% of the subjects did not show withdrawal symptoms 24–48 h after ibogaine's administration and did not seek the drug for at least 72 h. Withdrawal symptoms relief has occurred fastly, in about 1–3 h after ibogaine's administration.	Alper et al., 1999 [143]
Heroin and methadone	Report of the clinical experience of patients undergoing opiate detoxification, $N = 32$, inpatient, 12–14 days detoxification program.	Ibogaine HCl was administered at a dosage of 800 mg (approximately 10 mg/kg). Subjects also underwent motivational counseling and were forwarded to aftercare programs.	A single dose of ibogaine reduced withdrawal symptoms for up to 36 h, according to The Objective Opiate Withdrawal Scale and Heroin Craving Questionnaire scales. Depressive symptoms also decreased, according to the Beck Depression Inventory.	Mash et al., 2001 [144]
Alcohol, cannabis, cocaine, and crack	Retrospective study of patients treated in a private clinic, $N = 75$, 7 days outpatient/inpatient.	1–9 sessions of ibogaine (approximately 17 mg/kg) and cognitive behavioral therapy.	After the last session, all women participants remained abstinent, while 51% of the men were abstinent and another 15% were abstinent through other interventions. The abstinence lasted on average 5.5 months and 8.4 months after one session and multiple sessions, respectively. No serious adverse effects were reported.	Schenberg et al., 2014 [147]
Opioids (heroin and prescription opioids)	Retrospective chart review of participants admitted to a single residential ibogaine treatment center, $N = 50$, 1-week inpatient.	18–20 mg/kg ibogaine hydrochloride (oral administration).	After a one-week ibogaine detoxification protocol, the subjects showed a decrease in withdrawal and craving scores according to Clinical Opiate Withdrawal Scale, Subjective Opiate Withdrawal Scale and The Brief Self-Control Scale.	Malcolm et al., 2018 [145]
Opioids and cocaine	Open-label case series review, $N = 191$, 12 days inpatient.	8–12 mg/kg ibogaine HCl (oral administration) with motivational counseling and referral to aftercare programs.	The detoxification program with ibogaine reduced withdrawal symptoms, according to The Objective Opiate Withdrawal Scale. Ibogaine decreased craving scores assessed by Heroin and Cocaine Craving Questionnaires. Ibogaine was effective in decreasing withdrawal-related depressive symptoms.	Mash et al., 2018 [146]
Morphine	Randomized, double-blind, placebo-controlled prospective pilot study with noribogaine (ibogaine metabolite), $N = 27$.	Subjects on methadone substitution therapy received morphine orally a week before the intervention of a single oral dose of noribogaine (60, 120 or 180 mg) or placebo.	Noribogaine was well-tolerated despite ibogaine-like side-effects. The time taken by the subjects to order replacement therapy didn't differ between the treated and placebo groups. Noribogaine showed a tendency in reducing withdrawal symptoms, although the results were not statistically significant.	Glue et al., 2016 [148]
Ketamine				
Substance	Type of study and sample	Intervention	Main results	Reference
Alcohol	Controlled, randomized, $N = 186$, outpatient, 1-year follow-up.	Aversive therapy (Affective Contra-attribution method) with intramuscular administration of ketamine (3 mg/kg), while the control group received traditional methods of aversive therapy.	At the one year follow up, 69.9% of the group treated with ketamine was fully abstinent, compared to 24% of the control group.	Krupitsky et al., 1992 [167]
Alcohol	Controlled, $N = 211$, outpatient (data of 100 control patients and 111 from ketamine group collected in a period of 10 years of research).	Patients underwent a 3-month treatment course of rational, cognitive individual and group psychotherapy with the aim to establish a mental set of sobriety and a negative attitude to alcohol. Then, ketamine group received intramuscular administration (2.5 mg/kg) and psychotherapy and met for group psychotherapy the day after the ketamine session.	Abstinence of more than 1 year was observed in 65.8% of those who had undergone ketamine psychotherapy and 27.0% had relapsed, while only 24% of patients in the control group remained sober for more than 1y and 69% had relapsed. The rate of abstinence observed in ketamine group was 40.7% and 33.3% after 2 and 3 years, respectively, but the data was not collected from control group.	Krupitsky & Grinenko, 1997 [168]
Alcohol	Case reports of personal experience and data from about 70 patients treated with ketamine-enhanced psychotherapy in private practice. No controlled study.	Five different methods were employed. Ketamine doses were not clearly reported.	The average abstinence rate after one year varied between 25 and 70% according to the author, with the best result found for group residential treatment with repeated ketamine administration and increased intensity (using a revised exclusion criteria).	Kolp et al., 2006 [170]

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Table 2 (Continued)

Ketamine				
Substance	Type of study and sample	Intervention	Main results	Reference
Alcohol	Open-label study, $N = 5$, no control group, 4-week treatment; 4-week follow-up.	Combination of naltrexone (380 mg single dose) and ketamine (0.5 mg/kg intravenous infusion once a week for 4 weeks) in depressive patients with history of alcohol abuse.	Depressive symptoms reduced after the treatment and 4 of 5 patients reported improvement in alcohol craving and consumption.	Yoon et al., 2019 [171]
Alcohol	Randomized experimental study, single blind, $N = 90$, 9-month follow-up. The primary goal was to evaluate the effect of retrieval/detabilization of maladaptive alcohol memories on ketamine effect over the alcohol abuse outcomes.	Protocol consisted of three days: On day1 (baseline) and day 10 (post-manipulation) primary outcomes and questionnaire-based variables were assessed. On day 3 participants were subjected to memory retrieval procedure and 5 min later received a 30-min iv infusion of ketamine (350 ng/min) or placebo. A third group received ketamine without retrieval procedure.	Ketamine disrupted maladaptive reward memories in hazardous drinkers when administered immediately after their retrieval. The group which received ketamine following maladaptive reward memories intervention showed reduced reinforcing effects of alcohol and lower long-term drinking levels compared to the other groups (ketamine or retrieval alone).	Das et al., 2019 [176]
Alcohol	Randomized controlled trial (midazolam as control), $N = 40$, between-subjects, 5-week outpatient.	Motivational enhancement therapy, plus 52-minute intravenous administration of ketamine (0.71 mg/kg) or midazolam (0.025 mg/kg) at week 2.	Infusions were well tolerated. Patients from ketamine group presented longer abstinence and time to relapse, and reduced heavy drinking days compared with midazolam on the 21 days of monitoring. No significant difference was observed for craving, withdrawal and impulsivity. At the 6-month follow-up 75% of ketamine group self-reported abstinence.	Dakwar et al., 2020 [172]
Cocaine	Randomized experimental study (crossover, double-blind trial), $N = 8$, within-subjects, 9 days inpatient; 4-week follow-up.	10-min mindfulness-based exercise before 52-min infusion of ketamine (0.41 or 0.71 mg/kg) and lorazepam (2 mg - active control) starting on day 4; infusions separated by 48 h. The smaller dose of ketamine always preceded the highest dose.	Infusions were well tolerated. Ketamine significantly increased motivation for changing cocaine use and decreased cue-induced cocaine craving (change scores relative to pre-infusion) compared to lorazepam.	Dakwar et al., 2014 [182] (Qualitative analysis in Dakwar et al., 2014 [183])
Cocaine	Randomized experimental study, $N = 20$, 6 days inpatient study repeated three times with each hospitalization separated by 2 weeks; 2-week follow-up.	In each hospitalization, volunteers underwent the following protocol: washout period for two days; a session of free-base cocaine (25 mg) on day three; 52-minutes of saline infusion (first hospitalization) and ketamine (0.71 mg/kg) or midazolam (0.025 mg/kg) intravenous. infusion (second and third hospitalization, in counterbalanced order) on day 4; a 70-minute choice session on day 5; with discharge on day 6.	It was observed 67% of reduction in cocaine self-administration from baseline as compared to the control. Ketamine induced significant reductions in craving and cocaine use in the natural ecology relative to baseline, but the effect only differed from the midazolam group in the first follow-up.	Dakwar et al., 2017 [184] (Qualitative analysis in Dakwar et al., 2018 [185])
Cocaine	Randomized controlled trial, $N = 55$, 5 days inpatient and 5 weeks outpatient.	40-minute intravenous infusion of ketamine (0.5 mg/kg) or midazolam (0.025 mg/kg) on day 2 and discharge on day 5; Mindfulness-based relapse prevention therapy started at the first week and continued for 4 weeks, twice a week.	No serious adverse effects. Abstinence was maintained over the last 2 weeks of the trial in 48.2% of patients in ketamine group versus 10.7% in the midazolam group. The ketamine group was 53% less likely to relapse than the midazolam group and craving scores were 58.1% lower in the ketamine group. At the 6-month follow-up interview, 44% of the participants in the ketamine group self-reported abstinent, compared with none in the midazolam group.	Dakwar et al., 2019 [186]
Heroin	Randomized experimental study, $N = 70$, within subjects, inpatient (3–5 days) and follow-up to 24 m.	Intramuscular administration of ketamine 2.0 mg/kg (high dose) or 0.2 mg/kg (low dose, active placebo) plus existentially oriented psychotherapy.	High dose induced greater rate of abstinence within the first 24 m of follow-up than low dose. High dose produced a greater and longer-lasting reduction in craving. Both groups showed changes in depression, anxiety, anhedonia, and psychological profile assessed with a battery of verbal tests. However, many of the variables did not differ significantly between high and low dose groups.	Krupitsky et al., 2002 [187]

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Table 2 (Continued)

Ketamine				
Substance	Type of study and sample	Intervention	Main results	Reference
Opiates	Randomized experimental study (placebo-controlled, double-blind), <i>N</i> = 50, inpatient study; 4-month follow-up.	Patients under anesthesia received saline (control group) or ketamine infusion (0.5 mg/kg/h) preceding naloxone and naltrexone administration. Additional treatment with clonidine, carbamazepine, and clonazepam were delivered 48 h post-anesthesia, as well as a naloxone challenge test.	Ketamine infusion significantly suppressed the expression of precipitated opiate withdrawal and prevented rise in cardiovascular, respiratory, and neuroendocrine responses. Opiate withdrawal scores were significantly higher in control group, with patients from ketamine group requiring less additional carbamazepine and clonazepam to maintain the same level of opiate withdrawal symptoms during the first 48 h. Ketamine had no long-term effects on treatment of opiate dependence evaluated at 4 m follow-up.	Jovaiša, 2006 [189]
Heroin	Open-label study (Randomized, no placebo controlled trial), <i>N</i> = 59; inpatient (first session) and outpatient (second and third sessions); 1-year follow-up.	Single versus repeated (thrice) ketamine-assisted psychotherapy (2 mg/kg, intramuscular) plus two addiction counseling sessions (for both groups).	Abstinence was reported in 50% of the multiple ketamine-assisted psychotherapy group, compared to 22.2% in the single session group at 1-year follow-up. No differences between groups were found in craving, depression, and anxiety.	Krupitsky, 2007 [188]
Polysubstance abuse, including heroin and synthetic opioids	Case report. Two individuals with a history of polysubstance abuse, treating opiate withdrawal in the induction phase of buprenorphine.	Patient 1 received normal saline infusion and then ketamine (5 mg iv bolus, repeated once) plus magnesium sulfate and ketorolac (60 mg, iv), before receiving Suboxone® (buprenorphine/naloxone) sublingual film for 6 days. Patient 2 started the treatment with Suboxone® (2 sublingual tablets, later increased to 4 tablets, 2–3 times day) prior receiving ketamine (5 mg iv bolus, repeated once) and ketorolac (60 mg, iv).	Patient 1 reported relief of withdrawal symptoms and pain and three-month follow up found that patient was not experiencing any withdrawal symptoms. Patient 2 reported improvement of pain and two-month follow up revealed that patient was completely free of withdrawal symptoms.	Omoigui et al., 2011 [191]
Opioid	Case report. A woman who abused oxycodone and codeine as pain reliever.	Ketamine (1 mg/kg, oral solution) with gradual reduction of her opioid treatment (10% reduction in the initial dosage each two days).	The patient reported no withdrawal symptoms, pain or cravings while her opioid treatment was being reduced. Ketamine withdrawal was achieved without withdrawal symptoms after a few weeks.	Lalanne et al., 2016 [190]
Opioid	Open-label study, <i>N</i> = 3, outpatient, 2 to 3-week.	45 min infusion of ketamine (0.75 mg/kg), five transcranial magnetic stimulation sessions and trauma interventions using mindfulness based extinction and reconsolidation of memories over two weeks.	Craving was reduced in 65.7% and mindfulness was increased in 41.2% compared to baseline.	Pradhan & Rossi, 2020 [192]
Cannabis	Prospective pilot study (single-blind, no controlled), <i>N</i> = 8; 6-week outpatient.	Motivational enhancement therapy and mindfulness-based relapse prevention behavioral treatment plus 50-min infusion of ketamine (0.71 mg/kg) at week 2 with an optional 90-min infusion (1.41 mg/kg) at week 4.	The frequency of cannabis use decreased significantly. Participants related that their confidence regarding their ability to abstain from using cannabis in stressful situations was increased, although no significant differences were found on craving assessment.	Azhari et al., 2021 [193]
MDMA				
Substance	Type of study and sample	Intervention	Main results	Reference
Alcohol, cannabis, and tobacco, among other drugs	Reports of data gathered from individuals who received MDMA in a clinical setting, <i>N</i> = 28.	MDMA (75–150 mg, orally); optionally, a second dose of 50 or 75 mg was offered with the objective of prolonging the session. Some subjects received propranolol or tryptophan to reduce discomforts.	A decrease in substance use was reported for 14 subjects, while 3 reported an increase.	Greer & Tolbert, 1986 [208]
Alcohol	Prospective pilot study, <i>N</i> = 14, open label, within-subject, 8-week treatment, 9-week follow-up.	8-week detoxification protocol, with MDMA-assisted therapy performed on weeks 3 and 7 using the doses 125 and 62.5 mg orally two hours apart. On the sessions that MDMA-assisted therapy was not performed, subjects underwent psychotherapy sessions.	Alcohol intake average was decreased after one month and the reduction was maintained after 9 months. Also, the MDMA-assisted therapy promoted a reduction in anxiety and depression rates. No abnormal physiological effects were reported.	Sessa, 2021 [209]

ergic and mesocorticolimbic associated areas, including VTA, substantia nigra, and PFC, whereas it increased the GDNF expression in VTA and pro-BDNF expression in the NAc [135]. Similar effects on GDNF expression and ethanol self-administration were observed with noribogaine [136].

As expected, preclinical models have highlighted the potential of ibogaine to treat SUD, due to its multiple mechanisms of action. Ibogaine was demonstrated to decrease ethanol intake in three strains of ethanol-preferring rats. However, this effect was observed only in intraperitoneal and intragastric administration routes of ibogaine, but not subcutaneously, suggesting that the effects require the involvement of active metabolites since the formerly mentioned routes depend on the first-pass metabolism [137]. Additionally, ibogaine decreased morphine and cocaine self-administration in rats in a dose-dependent manner, being these effects dissociated from the tremorigenic effects of ibogaine. The authors pointed out that these anti-addictive effects observed may be associated with a decrease in DA and its metabolites levels in the NAc and striatum [138,139]. Another study showed that ibogaine attenuated morphine withdrawal-related behaviors in mice, including naloxone-induced jumping and NMDA-induced jumping in morphine-dependent and naive mice, respectively. This study suggests that ibogaine exerts its therapeutic effects on opioids withdrawal syndrome and acts as an NMDA receptor antagonist, as demonstrated by binding assay [140]. Other behavioral models showed that a pre-treatment of rats with 40 mg/kg ibogaine attenuated the CPP elicited by morphine or amphetamine, being these effects reported after one conditioning trial, but not after four trials spaced by a week, suggesting that continuous administration of ibogaine can lead to tolerance mechanisms [141,142].

The potential of ibogaine to decrease withdrawal symptoms during detoxification was also demonstrated in clinical studies. Alper et al. [143] conducted a retrospective study with 33 patients who met the DSM-IV criteria for OUD with physiological symptoms between 1962 and 1993 and were treated with ibogaine at a dose of 19.3 ± 6.9 mg/kg. Withdrawal symptoms were assessed by physicians and the individuals were monitored for 72 h. Ibogaine attenuated withdrawal symptoms in about 1–3 h and 76% of the patients had no symptoms after 24–48 h, whereas after 72 h none of the subjects consumed substances [143]. Nevertheless, another study evaluated 32 subjects who self-reported heroin or methadone dependence and met the DSM criteria for OUD. These patients underwent a detoxification program of 12–14 days and had their withdrawal and depressive symptoms assessed before and after an ibogaine intervention in a dosage of 800 mg (approximately 10 mg/kg). A single dose of ibogaine reduced the depressive state and decreased withdrawal symptoms for up to 36 h after the administration, as reported in the Objective Opiate Withdrawal Scale and Heroin Craving Questionnaire scales [144].

A retrospective open-label study assessed withdrawal symptoms in 50 subjects with OUD 24 and 48 h before and after a one-week detoxification protocol with ibogaine at doses of 18–20 mg/kg. The baseline evaluation showed that the patients had mild to severe withdrawal symptoms and after the one-week intervention protocol, the individuals showed a significant decrease in the related symptoms [145]. An observational open-label study with 191 patients presenting opioid and cocaine withdrawal symptoms according to the ASI and the DSM-IV showed that ibogaine detoxification protocol decreased the depressive symptoms and the craving scores of the individuals, according to the Heroin and Cocaine Craving Questionnaires [146]. Another retrospective study evaluated 75 subjects who reported alcohol, cannabis, cocaine, and crack use and were treated with ibogaine and Cognitive Behavioral Therapy jointly between 2005 and 2013 [147]. The individuals underwent 1 to 9 sessions and after the last session, all women participants were abstinent, while 51% of the men remained abstinent and another 15% were abstinent through other interventions. The abstinence lasted a median of 5.5 months after the first session, while for those who needed more sessions, the median before the first session and all subsequent sessions was 8.4 months. The authors emphasized

that compared with psychotherapy alone or topiramate pharmacological therapy, this result showed higher abstinence rates [147]. The first double-blind, placebo-controlled study with noribogaine was conducted by Glue et al. [148]. Twenty-seven patients in methadone substitution therapy started receiving morphine orally a week before the experiment and were treated with a single dose of noribogaine (60, 120 or 180 mg) or placebo. The time taken by patients to order replacement therapy did not differ between the noribogaine-treated and placebo groups, perhaps due to a study bias caused by the study design, according to the authors. Noribogaine showed a tendency to reduce withdrawal symptoms, especially with the dose of 120 mg, although this reduction was not statistically significant [148]. The study also showed that noribogaine was well-tolerated at the tested doses, despite some side effects, such as nausea, headache and visual changes. However, the authors suggest that electrocardiogram monitoring is required due to the QT interval cardiac effects that appear to occur, analogous to ibogaine.

Preclinical studies also evaluated the effect of noribogaine instead of ibogaine. Noribogaine at a dose of 40 mg/kg was effective in reducing morphine and cocaine self-administration in rats, and also antagonized the motor stimulant effect triggered by morphine and reduced dopamine and its metabolites levels in the NAc and striatum [149]. In addition, noribogaine dose-dependently decreased withdrawal symptoms in a naloxone-precipitated morphine withdrawal model in mice, while did not present CPP establishment, indicating that its anti-addictive effects do not work as a substitution therapy [150]. Nevertheless, noribogaine decreased in a dose-dependent manner the nicotine intravenous self-administration in a rat model, but at the highest dose moderately interfered with food intake, suggesting that it is capable of acting on non-drug reinforcing stimuli as well [151]. The authors suggested that the mechanism of action underlying nicotine reinforcing attenuation occurs due to $\alpha 3$ cholinergic nicotinic receptor inhibition, more specifically in the habenula.

Given that ibogaine produces effects that can limit its clinical application, such as tremors and arrhythmias [133], synthetic analogs are being developed in an attempt to disentangle the therapeutic effects of interest from unwanted effects. 18-MC is the main synthetic analog of ibogaine and has been shown to be as effective as ibogaine in reducing cocaine, morphine, and nicotine use in self-administration paradigm models [152]. Additionally, 18-MC demonstrated a decrease in ethanol intake in a strain of ethanol-preferring rats [153] and reduced the consumption in the binge-drinking behavior of C57BL/6 mice at a dose of 30 mg/kg [154]. In contrast to ibogaine, 18-MC has a selective effect on nicotine intake, and it has been hypothesized that its mechanism of action occurs primarily via the antagonism of the $\alpha 3\beta 4$ cholinergic nicotinic receptor and the consequent reduction of dopamine release in the medial habenula [153,154]. More recently, novel synthetic analogs of ibogaine have been developed and have shown promising results with safer profile in preclinical models, such as XL-008 and tabernanthol [155,156]. XL-008 improved the release of GDNF in C6 glioma cells, indicating a potential against SUD [156], while tabernanthol decreased alcohol- and heroin-seeking behavior in the two-bottle paradigm [155].

3.2.2. Ketamine

Ketamine was synthesized in 1962 by Calvin L. Stevens with the objective of developing an anesthetic that could be a reliable alternative to phencyclidine [157]. Ketamine presents a safe and effective profile for clinical anesthetic use due to its profound analgesic effects, safety as a short acting anesthetic and minimal side effects compared to phencyclidine, but the drug induces a state of disconnection from the environment, which led to the coining of the term “dissociative anesthetic” [157].

Ketamine is an arylcyclohexylamine with two enantiomers, (R)-ketamine and (S)-ketamine [158]. The drug is a potent, non-competitive NMDA receptor antagonist but it also affects several other cellular and molecular pathways [159]. It is a lipid soluble compound usually administered as an injectable solution either by intravenous or intramuscular

route, producing its effects on the central nervous system within 5 min after injection. Ketamine is a schedule III drug, meaning that despite its medical use, it may present great potential for abuse and dependence in humans [158,160]. Doses from 1 to 4.5 mg/kg of racemic ketamine are usually employed for induction of anesthesia with lower doses for maintenance [161], while subanesthetic doses are used for psychedelic experience [159]. On the other hand, very low doses are used on the treatment of depression and these doses generally do not produce strong psychomimetic effects that may underlie ketamine's rewarding properties [162].

In fact, several studies have suggested that ketamine is an effective drug for the treatment of major depression disorder, including treatment-resistant depression and to decrease suicidal thoughts [162–165]. Ketamine seems to produce fast and sustainable effects for up to two weeks after a single dose and it was hypothesized that the antidepressant effect is due to improved PFC glutamate homeostasis [166]. On the other hand, repeated low-dose ketamine infusions have shown to be more effective in treatment-resistant depression [162].

Regarding the use of ketamine for SUD, we found 16 clinical studies which evaluated its use for cocaine, heroin, and alcohol use disorder, and for the management of withdrawal symptoms from these substances, being two RCT, 5 RES, one prospective pilot study, three 3 open label studies, two case reports and three not classified (reports of data collected from several studies or treatments) (Table 2). Evgeny Krupitsky was a pioneer in studying the effect of ketamine in the treatment of drug dependence. His experience treating AUD patients was described in an anecdotal report for the Multidisciplinary Association for Psychedelic Studies [167] and in a review of ten years of research [168]. Krupitsky reported that from 111 alcohol users that received Ketamine Psychedelic Therapy, 65.5% showed total abstinence for more than one year, compared to 24% of the conventional treatment group. Ketamine Psychedelic Therapy (also referred to Ketamine-assisted Therapy) consisted of three phases [159,169]: a preparation session (when patients underwent a preliminary psychotherapy session with a psychotherapist that explained what are expected from the ketamine experience); the ketamine session per se (when the psychotherapist also interacted with the patient); and a group psychotherapy performed with patients and therapist the day after the ketamine session (whose goal was to help the patients to integrate insights of psychedelic experience into their lives). Krupitsky et al. [167] also reported the results of the combination of ketamine (3 mg/kg, im) and aversive approaches in alcoholism treatment, what they called “Affective Contra-attribution” method. The control group received traditional methods of treatment (not clearly explained in the paper). After one year, almost 70% of the experimental group treated with ketamine was fully abstinent from alcohol, compared to only 24% of the control group. However, this study does not present detailed experimental data to allow independent conclusions.

Kolp et al. [170] collected data from AUD patients treated with ketamine using different methods, based on various case reports and previous works of Krupitsky. The methods included individual outpatient treatment for 6 weeks and group psychotherapeutic programs offered as part of an intensive residential rehabilitation program with different durations (1 to 3 weeks), being the 3-week program composed by two administrations of ketamine. The average abstinence rate after one year varied between 25 and 70% depending on the treatment schedule. Although results seem very satisfactory, Kolp did not conduct formal clinical studies; instead, he collected data from patients from his private practice and the doses of ketamine were not reported, making the results questionable.

Recently, a small 8-week open-label pilot study evaluated the combination of naltrexone and ketamine in depressive patients with history of AUD [171]. Five patients received injectable naltrexone (380 mg) and 2–6 days later repeated intravenous ketamine treatment (0.5 mg/kg once a week for 4 weeks). The combination of naltrexone and ketamine was associated with reduced depressive symptoms and improvement in al-

cohol craving and consumption, measured by the Obsessive-Compulsive Drinking Scale.

So far, we found two controlled clinical trials with ketamine for AUD. Dakwar et al. [172] investigated the effect of a single ketamine infusion combined with Motivational Enhancement Therapy for AUD. In this randomized pilot trial, 40 patients received ketamine (0.71 mg/kg) or midazolam (0.025 mg/kg, active control) during the second week of a 5-week outpatient regimen. After 3 weeks of the infusions, 59.1% of participants in the control group had used alcohol with 40.9% reporting a heavy drinking day, while in the ketamine group the percentages were 47.1 and 17.6%, respectively. The participants were contacted and interviewed by telephone 6 months later, and among those that could be reached, 75% of the ketamine group self-reported being abstinent versus 27% in the midazolam group. The authors suggested that a single ketamine infusion improved measures of drinking in alcohol dependents, but they state that the data is preliminary and further research should be conducted in a large sample. As a secondary analysis, authors observed that ketamine led to a significant reduction in at-risk drinking and greater mystical-type effects and dissociation compared to the active control, increasing evidence to the importance of mystical-type experiences in SUD treatment [173]. The other randomized, double-blind, placebo-controlled clinical trial is being conducted as a preliminary assessment of the effectiveness of a combined repeated ketamine and psychological therapy program [174]. The protocol includes either three sessions of ketamine (0.8 mg/kg) or placebo (50 ml saline 0.9%) administered intravenously over 40 min plus either seven sessions of Manualized Psychological Therapy or an Alcohol Education Control. The final data of this study is yet not available, but a subsample with 12 participants that completed the trial was qualitatively evaluated by semi-structured interviews [175]. Authors report that ketamine treatment led to a significant change in relationship with alcohol and the set and setting was found to be important in determining acute ketamine experiences.

Das et al. [176] investigated whether intravenous ketamine given during the so-called “reconsolidation window” would interfere with the reconsolidation of maladaptive reward memories in harmful drinkers (people with high use of alcohol, but who do not have a formal diagnosis of AUD). Ketamine infusion (350 ng/mL for 30 min) following the retrieval/destabilization of maladaptive alcohol memories induced a rapid reduction in the reinforcing and motivational properties of alcohol and substantially lasting reductions in drinking levels compared to the other groups (see details of the protocol in Das et al. [176]). This result indirectly supports the idea that psychotherapeutic interventions are necessary to obtain the therapeutic effect of ketamine in the treatment of AUD/SUD. Other incomplete and ongoing studies evaluating the potential of ketamine for AUD are presented in Worrel & Gould [177].

The neurobiological mechanisms underlying the therapeutic effects of ketamine for AUD and some issues regarding its repeated use were discussed in a recent review [160]. It is known that chronic use of alcohol increases glutamatergic transmission through NMDA receptors potentiation in reward-related brain regions such as the NAc and dorsomedial striatum. Although acute administration of ketamine blockades NMDA glutamatergic receptors, studies suggest that repeated ketamine exposure enhances the glutamatergic transmission in NAc and other areas. Because alcoholism is a lifelong disorder, repeated ketamine infusions might be required for maintaining long-term alcohol abstinence, which brings concerns about the safety of its use in AUD [160].

Some retrospective studies analyzed the data of patients who received ketamine as adjunctive medication to benzodiazepines, specifically for management of alcohol withdrawal syndrome [178–180]. The authors showed that ketamine was well tolerated at low doses and appeared to reduce benzodiazepines requirements, i.e., the dose required for management of alcohol withdrawal syndrome. These studies analyzed only the data from the hospitalization period, and they did not report long term outcomes or abstinence and relapse rates. Krystal et al. [181] observed that recently detoxified alcohol-dependent pa-

tients showed reduced perceptual alterations, dysphoric mood, and impaired executive cognitive functions induced by ketamine compared to the healthy group, suggesting that alcohol dependence produced some alterations in NMDA receptor function possibly relevant for the treatment of AUD.

The effectiveness of ketamine was also investigated for cocaine use disorder. Dakwar et al. [182] evaluated the effect of ketamine infusion on cocaine users in a crossover, double-blind trial using lorazepam as an active control. Participants underwent a 10-minute Mindfulness-based Exercise and then received infusions of two doses of ketamine (0.41 and 0.71 mg/kg) or lorazepam (2 mg) separated by intervals of 48 h. The motivation to halt drug use and craving induced by drug cues were assessed 24 h after each drug administration. According to the authors, Ketamine infusions were well tolerated, increased the motivation to stop using substances and reduced cue-induced craving. Authors also evaluated the mystical-type effects (adapted from Hood's Mysticism Scale) and found that ketamine led to significantly greater acute mystical-type effects compared to the active control lorazepam and with higher scores obtained with the dose 0.71 mg/kg [183]. The authors suggested that psychological mechanisms may be involved in the therapeutic benefits of ketamine for SUD. In another study, Dakwar et al. [184] aimed to evaluate the effect of ketamine in the behavioral shifts in the salience of cocaine versus money paradigm in non-depressive cocaine users not seeking treatment. The authors reported that Ketamine infusion (0.71 mg/kg) decreased cocaine choice by 67% in post-baseline evaluation compared to midazolam (0.025 mg/kg) [184]. Similarly to the previous study, the authors found that mystical-type effects were greater after ketamine infusion compared to midazolam [185]. The same group also evaluated the effect of ketamine combined with Mindfulness-based Relapse Prevention Therapy [186]. Fifty-five individuals seeking treatment for cocaine use disorder were hospitalized for 5 days and received daily sessions of the behavioral treatment and a 40-minute intravenous infusion of ketamine (0.5 mg/kg) or midazolam (0.025 mg/kg) on day two. Participants returned twice weekly for 4 weeks for Behavioral Therapy sessions, monitoring and measures. About 48% of the participants from the ketamine group maintained abstinence after two weeks of the trial and only 10% in the midazolam group. Also, the relapse ratio and craving scores were statistically lower in the ketamine group compared to the active control group.

Other studies investigated the use of ketamine in the management of withdrawal symptoms of opioid dependents and detoxified patients. Krupitsky et al. [187] evaluated the effect of Ketamine Psychotherapy in heroin users that had just completed acute detoxification. The patients received Existentially Oriented Psychotherapy in combination with an intense hallucinogenic dose of ketamine (2.0 mg/kg, im), or a low, non-hallucinogenic dose (0.2 mg/kg im), employed as active placebo. The study was double-blinded, and the therapy consisted of a preparation session before the infusion and a sequent session to help subjects to establish a connection between the ketamine experience and their desire to quit substance use. A higher rate of abstinence as well as greater and longer-lasting reduction in craving were observed after the first two years of follow-up for the group that received the higher dose of ketamine, which also had greater positive change in nonverbal unconscious emotional attitudes. In a subsequent similar study, Krupitsky et al. [188] compared the effect of a single dose versus repeated sessions of Ketamine-assisted Psychotherapy in detoxified heroin users. All subjects had a ketamine (2.0 mg/kg, im) session prior to their discharge from the hospital and then were randomized into two groups: one received two Addiction Counseling Sessions and ketamine administration one and two months later, while the other had only the two Addiction Counseling Sessions. After a one-year follow-up, fifty percent of the subjects from the three sessions of Ketamine-assisted Psychotherapy group remained abstinent versus 22% from the control group, but no differences were observed in craving rates for heroin.

Another study investigated the effect of placebo (saline) or ketamine infusion (0.5 mg/kg/h) on the precipitated opiate withdrawal induced

by naloxone and naltrexone administration under anesthesia in patients with OUD. Ketamine significantly suppressed the expression of precipitated opiate withdrawal and prevented significant rise in cardiovascular, respiratory and neuroendocrine response compared to the control group [189]. Patients were enrolled in an aftercare program of their choice following hospital discharge and were contacted after 4 months when they answered a questionnaire based on ASI. Authors concluded that ketamine was an effective adjuvant in the acute phase of precipitated opiate withdrawal, but the 4-month follow-up showed that it had no long term effects on treatment of OUD. We also found some case reports about the use of ketamine in the management of opioid withdrawal symptoms. These reports suggest that ketamine was useful to reduce the severity of the withdrawal phase of patients under opioid detoxification [190,191].

An open-label proof of concept study was designed to test the feasibility and efficacy of ketamine (single infusion of 0.75 mg/kg administered over a 45-minute period), five Transcranial Magnetic Stimulation sessions, and trauma interventions using Mindfulness Based Extinction and Reconsolidation of Memories in three patients diagnosed with OUD [192]. The combination therapy reduced craving in 65.7% compared to baseline and increased mindfulness in 41.2% suggesting a synergistic effect of these therapies. No further clinical studies were found with ketamine for OUD. Another proof of concept trial evaluated the feasibility of ketamine in 8 cannabis-dependent individuals receiving Motivational Enhancement Therapy and Mindfulness-based Relapse Prevention behavioral treatments [193]. Participants received an infusion of ketamine (0.71 mg/kg) and a second infusion (1.41 mg/kg), in the case of non-responders. Self-reported cannabis use was decreased in the week after the first infusion, compared to baseline, and remained reduced at the end of the study.

The therapeutic potential of ketamine for SUD is supported by several preclinical studies. For instance, ketamine dose-dependently reduced ethanol drinking in alcohol-preferring rats in a model of self-administration [194] and reduced self-administration of nicotine in an operant box [195]. Franco et al. [196] evaluated the early-life ketamine exposure (from day 20 to day 39 postnatal) on the preference for ethanol in adolescent Sprague-Dawley rats. Ketamine administration in male rats attenuated the rewarding properties of ethanol in the CPP paradigm with all tested doses, while ketamine-pretreated female rats needed a higher dose of ethanol to exhibit ethanol-induced CPP. Fuller et al. [197] showed that D-amphetamine withdrawal decreased BDNF levels throughout the limbic system and these effects were reversed by treatment with ketamine on day 4 post-amphetamine, suggesting that ketamine could offer relief from withdrawal symptoms. Garcia-Carachure et al. [198] evaluated the effect of repeated ketamine treatment in juvenile and adult C57BL/6 mice from both sex in rewarding properties of sucrose and cocaine in adulthood. The results varied depending on the protocol, but in brief, they found that adolescent ketamine exposure increases later life preference for the rewarding properties of sucrose and cocaine in a sex- and age-specific manner. Other studies examined the effect of the enantiomers. (R)-ketamine attenuated naloxone-precipitated withdrawal signs in rats sub-chronically treated with morphine and blocked morphine-induced place preference in mice [199]. The authors also found that (R)-ketamine did not induce CPP by itself. Another study using the CPP model observed that (R,S)-ketamine reduced anxiety-like behavior of mice in an elevated plus-maze following morphine conditioning on CPP and prevented morphine-induced CPP [200]. In addition, ketamine also attenuated sucrose-induced CPP suggesting that besides alleviating negative affective states, it also may act by potentially impairing memory of morphine-context associations.

Finally, it is worth to mention that S(+)-ketamine presents more affinity for the NMDA receptor and it is postulated that it promotes greater analgesic and anesthetic activity with fewer psychotomimetic effects than the R(-)-ketamine form or the racemic mixture, providing similar clinical benefits with fewer adverse effects [158]. However, other studies suggest that (R)-ketamine might provide a better overall phar-

macological profile for the treatment of depression and SUD with less detrimental side-effects [199,201,202]. In conclusion, there is robust evidence of the therapeutic potential of ketamine for depression, but the potential benefits of each enantiomer, particularly for the treatment of SUD, still needs further investigation.

3.2.3. MDMA

3,4-methylenedioxyamphetamine, also known as “ecstasy”, “molly”, “XTC” or “Adam” is a psychedelic derived from amphetamine which was first synthesized in 1912 and subsequently patented by Merck in 1914. The increasing reports of use of this psychedelic in recreational context in the decades of 1960 and 1970 in the USA led to its classification as a Schedule I drug in 1985 [203,204]. MDMA is a ring-substituted amphetamine, and its action mechanisms are mainly due to the monoamines reuptake inhibition, more specifically serotonin, and to a lesser extent, norepinephrine and dopamine. MDMA also increases the release of monoamines and acts as a direct serotonergic agonist, binding with 5-HT_{1A}, 5-HT_{1B} and 5-HT_{2A} receptors, as well with adrenergic, histaminergic, muscarinic and sigma-1 receptors [203,205].

Frequently, MDMA use is associated with prosocial effects, including improved mood, friendliness and increased energy, which can be associated with oxytocin enhancement triggered by its action on 5-HT_{1A} receptors [203,206]. However, MDMA presents side effects, such as anxiety, arrhythmias, cognitive impairment, hyperthermia and serotonin syndrome risk and its metabolites can lead to liver toxicity and neurotoxicity [203,206]. Furthermore, it is also known that MDMA is commonly used jointly with other substances, especially alcohol and cannabis. When combined with alcohol, the MDMA reinforcement effect is increased due to the improvement in DA levels, leading to a euphoric state, and if combined with cannabis, its stimulant effect may be decreased [207].

Although MDMA-assisted Therapy has been studied for the treatment of psychiatric disorders such as post-traumatic stress disorder, anxiety and depression, its use for SUD remains poorly explored – we only found one prospective pilot study and one study with the objective of evaluating the psychological improvement of participants after a MDMA-assisted session (Table 2). This study describes the subjective reports of individuals with different psychological problems after having a MDMA session (with an optional second dose) and shows that besides having favorable effects on participants' attitudes and feelings, MDMA decreased substance use, including alcohol, cannabis and tobacco [208]. Sessa [209] hypothesized that MDMA can present therapeutic potential in AUD considering the trauma-related comorbidities that often follow this disorder and MDMA's empathogen features. In addition, the author suggests that therapeutic properties for SUD observed for classic psychedelics are an indication that further investigation on MDMA might also show promising effects since these substances share similar action mechanisms [209]. Jerome et al. [210] argued that MDMA-assisted Therapy can be useful in SUD considering its prosocial effects and the changes in perception, mediated by oxytocin release. In addition, authors discuss that the lack of studies on MDMA is partly due to substance misuse being an exclusion criterion in clinical trials involving this psychedelic since some authors fear that it may be liable to harmful use; however, in a clinical trial's controlled environment it is preconized as safe [210].

Recently, the first clinical trial evaluating the safety and tolerability of MDMA-assisted Therapy in patients who met the DSM-V criteria for AUD was conducted [211]. Fourteen patients underwent an 8-week therapy protocol and received oral administration of MDMA (125 and 62.5 mg, respectively, two hours apart) at weeks 3 and 7. All participants underwent prior detoxification and had their levels of alcohol consumption, anxiety and depression assessed for 9 months. In addition, physiological parameters such as blood pressure, temperature and heart rate were assessed during MDMA-assisted Therapy sessions, and no significant alterations were reported, and medical support was not needed. Regarding alcohol drinking rates, the subjects reported a mean

of 130.6 units of alcohol consumption per week before the intervention, and a decrease was observed one month after detoxification, which continued after 9 months, when the average weekly consumption was 18.7 units. Reductions in depression and anxiety rates were also observed. However, this study was not blinded or placebo-controlled and had a small number of subjects [211].

A few preclinical studies have indirectly evaluated MDMA's therapeutic effects for SUD. It was reported that MDMA decreased, in a dose-dependent manner, THC withdrawal symptoms in mice when administered both acutely and chronically [212]. Also, MDMA administration increased 5-HT release in the PFC. Taken together, these findings suggest that MDMA attenuates abstinence in rats acting through a serotonergic pathway capable of attenuating withdrawal behavioral signs [212]. Besides that, MDMA induced CPP in rats but decreased ethanol intake, in contrast with other drugs, such as morphine, which induces CPP and increases ethanol intake. However, this interference in alcohol intake was not specific, since water intake was also affected and it was not long-term observed [213]. In addition, a preliminary report showed that both MDMA and its enantiomer R(-)-MDMA reduced cocaine self-administration in squirrel monkeys. Furthermore, R(-)-MDMA was also able to block amphetamine-induced CPP in mice [214].

Taken together, the preliminary data assessed from both preclinical and clinical studies suggest that MDMA-assisted Therapy is worth being investigated, especially for the treatment for SUD. In particular, the benefits of MDMA-assisted Therapy should be evaluated in large samples of patients with SUD associated with psychiatric conditions.

3.2.4. Salvinorin A

Salvia divinorum, commonly named “pastora”, “sage”, or simply “salvia”, is a plant of the mint family (Lamiaceae) with psychoactive properties when its leaves are consumed by chewing, smoking, or as a tea [215]. The Mazateca's culture used *Salvia divinorum* for centuries both for ritualistic and therapeutic purposes as a medicine to treat headaches, anemia, rheumatism, diarrhea, among others [215]. Its main psychoactive component is salvinorin A, a neoclerodan diterpene that acts differently from most of the substances mentioned in this article since it has no affinity for the serotonergic receptor 5-HT_{2A} [216].

Animal and in vitro research show that salvinorin A acts mainly as a selective agonist for *kappa*-type opioid receptors along with allosteric modulation of endocannabinoid receptors type 1 [217–220]. Interestingly, the activation of *kappa*-opioid receptors reduces synaptic dopamine in different brain regions, including the NAc [221]. In some ways, it is the opposite effect of stimulant substances like cocaine and amphetamine, which increase dopamine release in these regions [222].

Studies using salvinorin A or *Salvia divinorum* itself in humans are limited and relatively scarce. The most recent research regarding the therapeutic potential of salvinorin A is largely related to its anti-inflammatory and analgesic capacity in addition to gastrointestinal and mood disorders, neurological diseases and drug addiction [218,223], but no clinical studies were found for salvinorin A and SUD. Despite the dearth of studies in humans, the therapeutic potential of *kappa*-opioid receptor agonists against dependence of stimulant drugs has been explored in preclinical research [224]. Tomasiewicz et al. [225] identified that an experimental *kappa* agonist drug reduced the reward-related effects of cocaine and might improve symptoms of conditions characterized by increased motivation and hyperfunction of brain reward systems, such as mania and stimulant intoxication, observed after chronic cocaine use in rodents. Another study with rats showed the ability of 2 mg/kg salvinorin A to block the behavioral sensitivity (locomotor response) generated by cocaine ingestion [226]. Kivell et al. [227] also confirmed the potential of salvinorin A (2 mg/kg) and Mesyl Sal B (0.3 mg/kg, a potent and selective *kappa*-opioid receptor analogue of salvinorin A) in mitigating the behavioral effects of cocaine-induced hyperactivity and behavioral sensitization to cocaine.

Other studies carried out by Morani et al. [228,229] showed that salvinorin A (0.3 mg/kg) might attenuate cocaine-induced drug-seeking

and cocaine prime-induced reinstatement in self-administration models in rodents. Researchers noted that salvinorin A attenuated the expression of behavioral sensitization induced by cocaine and did not attenuate responses to natural rewards, such as sucrose. It was also possible to observe that the substance did not affect spontaneous locomotion, cocaine-induced stereotypy, cocaine-induced hyperactivity, or conditioned taste aversion, suggesting that the dose of 0.3 mg/kg salvinorin A is sufficient to produce attenuation of drug-seeking in mice and does not cause sedation or aversion [229]. However, salvinorin A induced pro-depressive effects, which may limit its therapeutic potential.

Potter et al. [230] investigated the effects of repeated administration of salvinorin A (2 mg/kg) on the reward-potential of cocaine, together with the activation of extracellular signal regulated kinase (ERK), cAMP response element binding protein (CREB), and immunoreactivity of c-Fos within the NAc. Repeated administration of salvinorin A decreased reward-potentiating effects of cocaine along with changes in the immunoreactivity of proteins, such as increased phosphorylated CREB and increased phosphorylated ERK. A study with non-human primates observed that salvinorin A at very-low doses administered by repeatedly intravenous injections (0.3–1 µg/kg each injection) decreased the choice for cocaine and remifentanyl (a potent, short-acting synthetic opioid analgesic drug) in a self-administration model. The researchers suggest that the *kappa*-opioid agonist acts as a punisher to consumption of drugs in operant choice procedures [231].

Finally, it is important to note that the acute activation of *kappa*-opioid receptors may produce anti-addictive effects by regulating DA levels in the brain [221]. Currently, there is an effort to improve the pharmacokinetics and tolerability of this compound, based on the structure of salvinorin A [232]. While work in this field is still under development, analogs with improved pharmacokinetic profiles are demonstrating anti-addiction effects. Details about the studies involving synthetic opioid agonists and analogs of salvinorin A to treat addiction can be found in a review conducted by Kivell et al. [232]. Although research is still in the early stages, these compounds, together with the ancestral use of “*pastora*”, hold promise for the future development of anti-addiction therapies.

3.2.5. Cannabis and tetrahydrocannabinol

Cannabis spp. is an Asiatic plant that seems to have accompanied humankind's peregrinations [233] and it is one of the oldest domesticated plants known for several purposes [234]. For millennia cannabis has been used for fiber, textile, nutritional, therapeutic, sedative and ritualistic purposes, among several other applications.

The therapeutic employment of preparations containing cannabis is observed alongside a large array of archeological evidence retrieved throughout history from early cultures and was likely developed together with its spiritual usage [235]. The role as a therapeutic alternative for harmful substance use had been mentioned in scientific literature from the 19th century, when Birch [236] described how the administration of a mixture containing *Cannabis indica* immediately acted as an “appetite stimulant and inductor of sleep” for two patients presenting “chloral addiction and opium poisoning”, respectively. Similar findings were observed 110 years later, when Labigalini et al. [237] published an observational study with crack-cocaine users who attempted to halt consumption by substituting it with smoked cannabis.

An expressive number of high quality cross-sectional studies presented in a systematic review [238] demonstrated a trend of people who use drugs using cannabis and their derivatives as a substitute for alcohol, heroin and other opioids, stimulants and other substances they may consider harmful. In fact, several studies have reported reduction or substitution of alcohol consumption for individuals using cannabis [239–244]. Additionally, a recent systematic review demonstrated how legally accessed cannabis may serve as a substitute for alcohol [245]. Cannabis use also seems to be a recurrent strategy among people who use drugs for reducing harmful drug use, to mitigate specific symptoms of opioid withdrawal [246–248], and to improve the quality of life of opioid users

[249,250]. Those are important findings from a public health and epidemiology perspective, since opioid users, especially those who inject, are among the most vulnerable population of drug consumers and have greater risk of being exposed to infectious diseases [2,251]

The main active compounds of cannabis are THC and cannabidiol (CBD) which interact with cannabinoid receptors type 1 (CB₁) and type 2 (CB₂). Isolation of Δ⁹-THC was only achieved in the second half of the twentieth century [252], followed by the relatively recent discovery of the cannabinoid receptors 24 years later [253]. Δ⁹-THC is the major psychoactive, pleasure-related compound in cannabis, and it is responsible for repeated consumption and for some therapeutic properties [24]. Also, Δ⁹-THC may promote subjective effects and disturb time, space and sensory perceptions, which explain why it may be considered the ‘psychedelic’ component in cannabis [25]. Despite that, the classification of cannabis as a psychedelic substance remains unsettled. Experiencing the psychedelic effects of cannabis may be seen by some authors [254] as a symptom linked to trait impulsivity, for instance, although a recent study demonstrated that high doses of cannabis can induce mystical experiences similar to those experienced with psilocybin [255]. Although milder than entheogens like mescaline and psilocybin, cannabis has also accompanied humankind and it is still sought in contemporary times for inducing mystical, transcendental, psychedelic or hallucinogenic experiences [256,257]. Additionally, psychedelic substances have diverse pharmacology and heterogeneous mechanisms of action [258] and are mostly considered as such by its ability to promote altered states of perception and thoughts, although the terms ‘classic psychedelics’ or ‘serotonergic psychedelics’ are being employed to elucidate these particularities [19,259]. Nonetheless, organizations such as the Beckley Foundation and the Multidisciplinary Association for Psychedelic Sciences hold important publications and ongoing research investigating therapeutic effects of cannabis and cannabinoids [260–262]. This session attempted to gather studies investigating how the psychedelic component in cannabis, THC, could act as therapeutic agent for continuous drug consumption or SUD.

Of all the several roles that the endocannabinoid system portrays, signaling on the brain reward circuitry might explain its conception as a promising target for the treatment of SUD. CB₁ receptors are extensively located in the brain and their modulation over striatal regions such as the VTA and NAc pathways regulate reward tone [263]. Cannabinoids, especially Δ⁹-THC, but also CBD and synthetic analogues, are being extensively studied as a potential cannabinoid-receptor agonist therapy for SUD, perhaps following the same trend of nicotine [264,265] and opioid [266] agonist and antagonist pharmacotherapies. Currently, there are three cannabinoid-based medications approved for the treatment of symptoms of cancer, chronic pain, and multiple sclerosis [267]. Dronabinol, a synthetic Δ⁹-THC; nabilone, a Δ⁹-THC analog; and nabiximols, a preparation of two different chemovars of cannabis oil with a 1:1 Δ⁹-THC and CBD ratio, are being tested for decreasing withdrawal symptoms severity, especially in cases of Cannabis Use Disorder (CUD), but also in cases of OUD and Tobacco Use Disorder. Sloan et al. [268] presented a high-quality review of RCT and within-subject randomized trials using these medications for those conditions, and an update of a Cochrane Systematic Review demonstrated that only preparations containing Δ⁹-THC were effective for decreasing the severity of withdrawal symptoms in patients with CUD [269]. We will consider these studies and gather newly published data to appraise this review of cannabis and cannabinoids for SUD.

We selected 19 studies demonstrating the role of Δ⁹-THC as a cannabinoid receptor agonist therapy for people diagnosed or demonstrating patterns of SUD (Table 3). Six of those studies are high quality RCT, one demonstrates the role of dronabinol in OUD, in addition to opioid-antagonist XR-naltrexone [270], and the other five used nabiximols [271–273] and dronabinol [274,275] for CUD. Levin et al. [275] was the only RCT that could not achieve most of their proposed outcomes, which authors attributed to an unexpected difficulty for participants tolerating the maximum doses of lofexidine, an adrenergic α_{2A}

Table 3
Human studies using cannabis and oral cannabinoids for substance use disorder.

Cannabis and oral cannabinoids				
Substance	Type of study and sample	Intervention and compositions	Main results	Reference
Cannabis	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 7, 35-day, 15-day inpatient, 5-day outpatient, 15-day inpatient.	Dronabinol 10 mg + smoked cannabis (3.04% THC).	Dro 10 mg decreased ratings of anxiety, trouble sleeping and chills during abstinence. Craving reported as “Mild” to “Moderate”. No effect on sleep ratings. Dro decreased performance on DSST. Dro 10 mg increased food intake and body weight average was higher during cannabis abstinence. No effect on tobacco smoking.	Haney et al., 2004 [277]
Cannabis	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 8, 40-day, outpatient.	Dronabinol 10, 30 or 90mg.	Reduced withdrawal symptoms severity. Significant effect on craving. Improved overall subjective sleep quality. Increase on body weight. Dose dependent THC effect, “drug effect/good effect/liked/high” ratings observed especially with 90 mg. No serious adverse events.	Budney et al., 2007 [276]
Cannabis	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 8, 28-day, inpatient phase / outpatient washout (7 days each phase).	Dronabinol 60 mg or Lofexidine 2.4 mg or the combination Dronabinol 60 mg + Lofexidine 2.4mg.	Dro 60 mg reduced withdrawal symptoms severity (anorexia, weight loss, sleep latency). Dro+Lofexidine decreased number of cigarettes per day. Recreation period for socialization was decreased by each medication. All three medication decreased systolic and diastolic blood pressure. Lofexidine and Dro+Lofexidine decreased cannabis self-administration. Dro+Lofexidine doubled the chances of achieving consecutive days of abstinence.	Haney et al., 2008 [278]
Cannabis	Randomized controlled trial, <i>N</i> = 156, 9-week, outpatient.	Psychotherapy + Dronabinol 40mg.	Reduced withdrawal symptoms severity. Higher retention in treatment. No abstinence score differences. Medication well tolerated.	Levin et al., 2011 [274]
Cannabis	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 11, 24-day, inpatient.	Nabilone 6 or 8 mg + smoked cannabis (5.6% THC).	Both Nabilone conditions decreased cannabis use and continued decreasing 3 days after. Decreased ratings of “irritable”, “bad effect”, “social” and “ <i>marijuana</i> craving”. Increased numbers of hours slept and of “sleep well” and reduced “woke often” ratings. Increased caloric intake, number of daily eating occasions and altered macronutrient intake. Reduced task performance on the DSST. No influence on tobacco consumption.	Haney et al., 2013 [283]
Cannabis	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 13, 5-day, outpatient.	Dronabinol 30, 60 or 120 mg.	Decrease of loss of appetite, diarrhea, nausea, stomach pain, irritability, chills and headaches. Cognitive performance 3% less accurate during Dro conditions than placebo.	Vandrey et al., 2013 [282]
Cannabis	Randomized controlled trial, <i>N</i> = 51, 6-day, inpatient.	Psychotherapy + Nabiximols (88.4 mg THC, 80 mg CBD).	Reduced withdrawal symptoms severity. Higher retention in treatment. Use reduction in follow-up, no score differences. Medication well tolerated. No serious adverse events.	Allsop et al., 2014 [271]
Cannabis	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 14, 3-day, inpatient.	Dronabinol 10 or 20mg.	Lowered blood pressure. Elevated heart rate. “Urge to smoke” / “Desire to use” attenuated. Anxiety decreased.	Lundahl & Greenwald, 2015 [281]
Opioids	Randomized controlled trial, <i>N</i> = 60, 6-week, inpatient.	Psychotherapy + Dronabinol 30 mg; XR-naltrexone up to 25 mg.	Withdrawal scores decreased. Similar retention rates. Cannabis smokers had lower insomnia and anxiety ratings. Medication well tolerated. No serious adverse events.	Bisaga et al., 2015 [270]

(continued on next page)

Table 3 (continued)

Cannabis and oral cannabinoids				
Substance	Type of study and sample	Intervention and compositions	Main results	Reference
Cannabis	Randomized controlled trial, <i>N</i> = 122, 8-week, outpatient.	Psychotherapy + Dronabinol 60 mg + Lofexidine 2.4mg.	No abstinence score differences. Withdrawal scores decreased over time. Similar retention rates. Medication not well tolerated. No serious adverse events.	Levin et al., 2016 [275]
Cannabis	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 11, 8-day, inpatient.	Nabilone 6 mg + Zolpidem 12.5 mg or Zolpidem 12.5mg.	Both medications combined: Reduced withdrawal symptoms severity. Decreased disruptions in sleep, mood and food intake. Decreased cannabis use. Increased rating of strong/good/bad drug effect. Neither medication altered cognitive performance.	Herrmann et al., 2016 [284]
Cannabis	Prospective pilot study, <i>N</i> = 9, 8-week, outpatient.	Nabiximols (108 mg THC, 100 mg CBD).	7 of 9 participants remained abstinent. Higher intake on fixed than self-titrated condition. Experiencing “high” was lower in Sativex than SAU. Withdrawal scores decreased. No significant craving scores. Decreased THC and higher CBD mean concentrations observed during both conditions.	Trigo et al., 2016 [286]
Cannabis	Open-label trial, <i>N</i> = 4, 3-month self-titration phase, outpatient.	Nabiximols (113.4 mg THC, 105 mg CBD).	No serious adverse events. Reduction of cannabis intake during intervention. No compensatory increases of other substances (tobacco, caffeine, alcohol). 1 of 4 subjects achieved abstinence and remained during the study.	Trigo et al., 2016 [287]
Cannabis	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 12, 10-week, outpatient.	Nabilone 2mg.	Follow-up average cannabis use was 17.9% days/week vs. 92.9% at baseline. Nabilone and placebo groups both self-reported use reduction. Medication deemed safe and tolerable. No serious adverse events.	Hill et al., 2017 [285]
Cannabis	Open-label trial, <i>N</i> = 12, 6-day open-label cannabis-smoking phase.	Bediol® (3.6% THC, 7.1% CBD) or Bedrobinol® (9.1% THC, <0.2% CBD), smoked in cigarettes.	Acceptance and appreciation of proposed cannabis, especially Bedrobinol for “stronger kick”. No side effects of smoking. No psychosis-like signs observed. Nausea and dizziness occasionally reported. No serious adverse events.	van Amsterdam et al., 2018 [288]
Cannabis	Randomized controlled trial, <i>N</i> = 40, 12-week, outpatient.	Psychotherapy + Nabiximols (113.4 mg THC, 105 mg CBD).	Decreased use in both groups. No abstinence score differences. No withdrawal scores differences. Craving scores showed time X treatment interaction. No serious adverse events.	Trigo et al., 2018 [273]
Cannabis	Randomized controlled trial, <i>N</i> = 128, 12-week, outpatient.	Psychotherapy + Nabiximols (88.4 mg THC, 80 mg CBD).	Decreased days of use. Similar retention rates. Medication well tolerated. Comparable improvements in health status. No serious adverse events.	Lintzeris et al., 2019 [272]
Opioids	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 12, 5-week, inpatient.	Dronabinol 5, 10, 20, 30 or 40 mg + Oxycodone 30 mg or 60mg.	Higher doses of Dro (20, 30 and 40 mg) increased anxiety and heart rate. Dro 40 mg was not tolerated.	Jicha et al., 2015 [279]
Opioids	Randomized experimental study (within-subject placebo-controlled), <i>N</i> = 12, 5-week, inpatient.	Dronabinol 5, 10, 20 or 30 mg + Oxycodone 30 mg or 60mg.	VAS: Dro 30 mg produced “drug dose” and liking” effects, and dose x interaction for “high”. Dro 20 and 30 mg suppressed opioid withdrawal scores and reduced ratings of symptoms (backache, feeling sick, muscular tension, runny or watery eyes, and rhinorrhea).	Lofwall et al., 2016 [280]

Dro = Dronabinol; Nbx = Nabiximols; SAU = Smoking as Usual condition; VAS = Visual Analogue Scale; DSST: Digit Symbol Substitution Test.

agonist, used as a concomitant pharmacological intervention in this RCT. Despite this exception, all of the interventions were well tolerated, and no severe adverse events were reported. All studies reported either decrease of withdrawal symptoms severity [270,271,274,275] or reduction in substance use and craving [272,273], and all of the studies had psychotherapy sessions such as Relapse Prevention, Motivational Enhancement Therapy or Cognitive Behavioral Therapy as additional interventions. Ten randomized experimental studies are within-subject placebo-controlled trials - seven studies used dronabinol as main intervention [276–282] and three used Nabilone [283–285]. We also included one prospective pilot study using nabiximols [286] and two open-label trials [287,288].

Interventions for opioid users were composed of dronabinol and oxycodone, a synthetic opioid [279,280]. Participants did not tolerate the highest dose (40 mg) of dronabinol [279], but lower doses (20 mg and 30 mg) suppressed opioid withdrawal scores and reduced ratings of symptoms like backache, nausea, lacrimation and rhinorrhea [280].

All other studies appraised cannabis users presenting patterns of CUD for their interventions. Haney et al. [278] proposed three medication conditions for inpatient and outpatient settings in a laboratory model of relapse: dronabinol 60 mg, lofexidine 2.4 mg, and dronabinol together with lofexidine, at the same dosages. Dronabinol alone reduced the severity of withdrawal symptoms such as anorexia, weight loss and sleep latency, and both lofexidine alone and dronabinol combined with lofexidine decreased cannabis self-administration. The other intervention with complementary pharmacotherapy was presented by Herrmann et al. [284] in which participants used nabilone (6 mg) and zolpidem (12.5 mg) or zolpidem alone at the same dosage in a laboratory model of relapse. The combination of both medications significantly decreased cannabis self-administration, reduced withdrawal severity, decreased disruptions in sleep, mood and food intake and decreased cannabis use while it did not alter cognitive performance. Hill et al. [285] used nabilone (2 mg) in a randomized prospective pilot study to assess safety, tolerability and efficacy for cannabis users but did not report treatment effects. Haney et al. [283] studied two doses of nabilone (6 mg or 8 mg) while participants also had access to smoked cannabis (5.6% Δ^9 -THC). Both nabilone conditions reduced cannabis use during the intervention and continued to do so during the three-day follow-up, increased caloric intake and the number of hours slept, but reduced cognitive task performance on the Digit Symbol Substitution Test. Haney et al. [277] also had active smoked cannabis (3.04% Δ^9 -THC) simultaneously with cannabinoid-based medication in an inpatient and outpatient design in which authors presented two studies, one tested divalproex, a mood stabilizer, and the other 10 mg of dronabinol, which was considered more effective than the former for lowering cannabis withdrawal. The dronabinol group reported craving to smoke as “mild to moderate”, as compared to “moderate to severe” in the placebo condition, had no effects on sleep ratings, and decreased performance on the Digit Symbol Substitution Test.

Some studies proposed three different conditions of intervention for participants: “smoking as usual” (SAU), where they smoke cannabis at their usual rate; “fixed”, a constant, determined dose of a medication (i.e., cannabinoids); and “self-titrated”, in which the patient could use the medication whenever they want, with a determined maximum dose. Trigo et al. [286] demonstrated that nabiximols effectively decreased withdrawal scores, and observed higher CBD than THC mean concentration in urinary and plasma assessments, and that there was a higher intake of cannabinoids during the fixed rather than the self-titrated condition. Scores of experiencing “high” were lower in both fixed and self-titrated nabiximols than SAU conditions. Budney et al. [276] proposed five days of abstinence conditions with three doses of either 10, 30 or 90 mg of dronabinol alternated with three SAU conditions lasting seven to nine days, and demonstrated reduction in withdrawal symptoms, significant effect on craving scores and sleep quality, and also an increase on body weight associated with dronabinol. Vandrev et al. [282] tested three doses of dronabinol (30, 60 and 120 mg), each dosage separated

by nine days of SAU condition, and reported dose-dependent attenuation of cannabis withdrawal symptoms and a slight 3% less accurate cognitive performance during dronabinol conditions than placebo. We retrieved one study that assessed cue-elicited effects of cannabis using a within-subject crossover design [281], with 10 and 20 mg of dronabinol, and it reports a decrease in “urge to smoke”, desire to use and anxiety scores. Two open label studies are also worth mentioning. Trigo et al. [287] evaluated a self-titration phase of nabiximols associated with behavioral psychotherapy and reported a follow-up average cannabis use of 17.9% days/week vs. 92.9% at baseline. Finally, to our knowledge, the only open-label trial of a cannabis-smoking phase studying harm reduction strategies was conducted by van Amsterdam et al. [326]. Researchers proposed to 12 diagnosed schizophrenic patients that consumed high potency cannabis to substitute the cannabis of their choice with two selected cannabis chemovars: Bediol® (3.6% Δ^9 -THC, 7.1% CBD) or Bedrobinol® (9.1% Δ^9 -THC, <0.2% CBD), and they attempted to assess appreciation and acceptance scores. Bedrobinol® had the highest levels of appreciation and no psychosis-like signs were observed with either chemovar, suggesting that it is possible to develop specific chemovars to appraise users of high-potency cannabis and serve as a harm reduction strategy, although further research is needed. No serious adverse events were reported in any of the aforementioned studies.

In this session we discussed high quality studies focusing on THC, considering that it may be one of the main compounds responsible for the psychedelic effects found in cannabis, though certainly minding the considerable differences between the isolated and even synthetic cannabinoid preparations and the *entourage* effect promoted by the consumption of whole cannabis derivatives [289,290]. Most studies demonstrate that medications containing THC, alone or together with CBD, are effective for attenuating withdrawal symptoms and reducing the consumption of cannabis, although positive results for opioid use were also observed. RCT with CBD, although not included in this review are also worth mentioning, as they present the prominent effectiveness in cases of use of opioid, stimulants and cocaine addiction [291–294].

Notably, high quality RCT using cannabis *in natura* are still scarce, as it is still a Schedule-I substance that prevents the advance of clinical studies. Several cross-sectional and also high quality longitudinal studies (e.g., Cunningham et al. [295]; Socías et al. [296]) are being conducted and are already providing consistent evidence aligned with RCT with isolated cannabinoids, that cannabis may be an alternative or complementary treatment for SUD, especially for OUD. However, the relation of complementarity and/or substitution with some substances (i.e., alcohol) is still being observed [297,298].

Similarly to psychedelics, some cannabis chemovars, specially THC-predominant profiles, may induce transformative experiences [255] and may substitute more harmful drugs [299], and although studies with isolated cannabinoids are advancing, higher quality studies using commercially-available, legally-obtained cannabis are necessary to understand this effects. As van Wel et al. [254] demonstrated in their study, experimenting with specifically selected chemovars may be an alternative to safely conduct high quality assessment and clinical studies as our understanding of the biochemical properties of cannabis advances [300] and we begin to tackle obstacles that isolated cannabinoids and the harms that uncontrolled high potency cannabis may present [301,302].

3.3. Online surveys and the use of micro dosing

We found publications of cross-sectional studies based on anonymous online surveys in which respondents self-report their experiences with various psychedelics, especially LSD and psilocybin, in non-clinical settings, as a strategy to cope with SUD. Such studies usually present results for all psychedelics included in the survey without clear specifications. Participants who used psychedelics reported decreases in the severity of withdrawal symptoms and craving, and cessation or reduction of alcohol use [303], cannabis, opioids and stimulants [304]. John-

son et al. [305] gathered information from individuals who had “quit or reduced smoking after a psychedelic experience”, and from a total of 358 participants, 38% reported quit smoking and 28% reported expressive reductions on smoking rates after psychedelic use, while 34% relapsed after temporary smoking reduction. The differences between groups of all substances were significant, with disadvantage for the relapse group. Lawn et al. [306] conducted a survey involving over 90,000 respondents to analyze well-being measures, harmful alcohol consumption and acute subjective substance effects in past-year ayahuasca users. The study reported greater well-being rates among ayahuasca users when compared to other classic psychedelics users (mainly LSD) and non-psychedelics substance-using respondents. Ayahuasca users also reported less harmful drinking than other classic psychedelic users, although these two groups reported greater harmful drinking than the non-psychedelics substance-using group. In another online survey with individuals who used various psychedelics at microdoses as a form of therapy, Lea et al. [307] found that 39% of respondents reported mental health or substance use problems as their primary motivations to use psychedelics. Improvement of mental health aspects were reported by 44% of all respondents and minor improvements by 35.8%, along with partial reduction of some medication and non-medical substance use. Another study observed self-reported improvement in psychiatric conditions, AUD and SUD by respondents who took mescaline in naturalistic context, and reported that these improvements were significantly associated with higher ratings of acute psychological factors including mystical-type experience and psychological insights [127].

Anonymous online surveys are a useful tool to gather information about use of psychedelics without the costs and bureaucracy of a controlled study. However, this type of study presents important limitations. Participant selection might exclude those who have tried psychedelics as an alternative treatment but got negative results; recall bias may lead to inaccurate responses; volunteer bias favors the report of positive outcomes; there is no access to the exact dosage used, their source and purity; many uncontrolled variables regarding the set and setting, which greatly influence the experience; and the absence of a control group.

Microdosing has been growing as another modality of psychedelics consumption among individuals dealing with mental health issues, and researchers are starting to investigate more about it. The placebo effect might also influence the results, as shown by Szigeti et al. [308] in a ‘self-blind’ trial in which participants were given online instructions on how to incorporate placebo control into their microdosing routine without clinical supervision. The results showed significant improvement for both the microdosing and the placebo groups, but no significant between-groups differences.

3.4. Limitations and reflections

This review aimed to discuss the therapeutic potential of classic and non-classic psychedelics for SUD. It is worth mentioning that although numerous, the quality of studies is still not standardized. For instance, while there are several clinical studies with LSD conducted in the years 1950–1970, most studies had no control groups or employed statistical analysis, while a few recent studies with psilocybin, ketamine and cannabis preparations adopted more rigid experimental designs. On the other hand, this is not a systematic review, and we did not employ statistical methods to evaluate the data. We must declare that despite our effort to search by clinical studies published with psychedelics and SUD, occasionally some studies were not available in the databases searched. Also, papers that the full text could not be obtained or published in other languages were not included.

Another important limitation refers to the methodological classification of the studies. The methods and experimental design of the papers were carefully examined in order to classify the type of study using contemporary scientific terminology (see Box 1). However, most of early studies do not fit in such classification. Indeed, clinical studies with psychedelics have some important particularities: First, psychedelics are

controlled substances, most of them on schedule I, which makes the approval of clinical protocols more difficult. Double blind studies are difficult to be designed because these substances produce evident psychoactive effects, so many researchers have employed an active placebo (usually another psychoactive substance or lower doses of the psychedelic under evaluation), but even following these protocols there is no guarantee of blinding, especially if the participant had a previous experience with the psychedelic and is familiar with the effects [309].

We also would like to point out some possible limitations in the experimental design of clinical studies. Several articles do not describe in detail information about the experimental design, such as randomization, allocation and blinding procedures. Several studies did not clearly report the dose, route of administration, time of treatment and experimental N. Most studies evaluated the outcomes for short periods (a few weeks or months) and some follow-up measures were mainly accessed by phone and reached only a small number of participants. In some studies, different attention was given for experimental and control participants in the follow-ups. Also, the criteria of inclusion and exclusion of participants is critical for the obtained results. It is well known that individuals looking for treatment are more likely to succeed than participants not interested in treatment, and are more likely to report improvement in subjective scales, which may lead to selection bias [2,310,311]. Muthukumaraswamy et al. [312] alert that high levels of response expectancy by participants may favorably interfere in the results of clinical studies and suggest that RCT with psychedelics should routinely measure de-blinding (unblinding) and expectancy. Mertens & Preller [309] also warn for the possibility of a nocebo effect in patients that realize they received the placebo or active placebo and became disappointed, which could produce negative responses. Finally, it is important to keep in mind that self-reported evaluations are subjected to individuals’ mood and are difficult to confirm. All these limitations suggest that the clinical results described here must be taken with prudence. Other important limitations are discussed in some opinion reviews and systematic reviews [28,29,31,33,36,309,313–316].

3.5. Main remarks and future directions

The treatment of SUD is very complex, and the results of pharmacological interventions are limited, with high relapse rates. Because heavy drug use may cause significant personal and social burdens, any treatment capable of potentially reducing the use of such substances deserves attention. Psychedelic substances have always aroused the interest of the scientific community, with several recent studies focusing on the investigation of their therapeutic potential for mental health disorders and SUD. Research on Psychedelic-assisted Therapy for SUD seem promising, especially for decreasing substance use, increase abstinence time and decrease the number of days of heavy use. Some psychedelics have demonstrated fast and long-lasting satisfactory effects with small to moderate doses, while other studies suggest that higher doses inducing mystical experience are necessary to produce the therapeutic effect. Also, some works suggest that repeated administration seems more effective to obtain perdurable effects. As noted [28], although the long-lasting effects may be very satisfactory for many patients, more work needs to be conducted to understand in which situation re-dosing would be desirable, for example, for patients in risk of relapsing.

Another important aspect of discussion is that psychedelic therapy frequently associates psychedelic use and one or more psychotherapy sessions. There is strong evidence that psychotherapy *per se* can result in beneficial outcomes for SUD, and the efficacy of the psychedelic treatment without therapeutic interventions is still a matter of debate. It seems that the association of psychedelic treatment and psychotherapy is the norm in most studies, but there are reports of people who use psychedelics without therapeutic supervision as a strategy to cope with SUD. The *set* and *setting* also seems to be fundamental influence on studies outcomes. Andersen et al. [28] report that some studies that

neglected the therapeutic setting and support generally failed to find positive effects.

Despite all the problems related to SUD, it is intriguing that the adherence to the current treatments is still low; consequently, it is important to open possibilities beyond our standard repertoire of treatment modalities. Recently, *in vivo* and *in vitro* studies have shown that psychedelics, including DMT, LSD, and ketamine can robustly increase neurogenesis and/or spinogenesis, mechanisms that are important for neuronal plasticity and are related to SUD behavior [30,317]. It was suggested that psychedelics allow the reorganization of disordered neural pathways in the default mode network and also attenuate maladaptive signaling in mesolimbic reward pathway [31]. More preclinical studies are necessary to evaluate the molecular basis of each psychedelic mechanism of action, as well as robust clinical studies with large samples to confirm the potential of psychedelics for SUD. Additionally, there is great potential in the use of psychedelics as part of a therapeutic strategy to cope with comorbidities associated with SUD such as depression, anxiety, obsessive-compulsive disorder and post-traumatic stress disorder [23,35,76,107,163,318] which should also involve adequate therapeutic support. Future clinical studies should always include the most suitable control groups and strategies to guarantee high quality study designs, with validated scales and instruments, appropriated follow-ups and rigorous statistical analysis.

In conclusion, clinical studies suggest that ketamine may be effective for AUD, reduction of cocaine craving and management of opiate withdrawal, and preliminary studies indicate that psilocybin may be effective for AUD and tobacco use disorder. THC and other oral cannabinoid preparations seem to be effective for CUD and possibly OUD, while ibogaine, used in detoxification clinics, seems useful to avoid withdrawal symptoms and craving mainly for opioids, alcohol and cocaine. Based on observational studies, the use of ayahuasca in ritual context seems to help individuals to cope with SUD. However, current literature does not support the use of LSD for SUD and there are no controlled clinical studies with mescaline, MDMA and salvinorin A for SUD. Importantly, in most controlled clinical studies, the beneficial effects observed were partial and temporary and some sort of psychotherapy seems to be necessary.

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Box 1. Glossary for clinical, experimental and observational studies.

Between-subjects design — A design of intervention in which different participants are exposed to a single condition (i.e., Group A takes medication, Group B takes placebo).

Case reports — Studies that report the development of a case registered by the authors. Usually involves a small number of participants and presents only a descriptive analysis.

Clinical trial — Studies conducted with humans to evaluate a medical condition. Usually controlled with placebo with a randomized sample.

Cross-sectional — A type of design of observational studies in which data from a population is analyzed at a specific time point, used for surveys assessing the prevalence of a condition (i.e., drug use self-report from residents of a country).

Fixed condition — A condition in which participants take their medication (or placebo) at a fixed dose, (i.e., three puffs from a spray in the morning, three in the evening).

Inpatient — Studies in which participants must spend at least one night properly accommodated in a hospital or clinical setting.

Longitudinal — A type of design of observational studies in which data is collected over a period of time from the same sub-

jects, to detect changes in characteristics of a targeted population (i.e., effects of five years of alcohol use in men).

Observational studies — Studies that are not experimental and provide qualitative evidence.

Open label trial — A type of design of study in which both participants and research team are aware of the intervention and no placebo is used.

Outpatient — Studies in which participants do not spend the night in the hospital, clinical or laboratory setting and are dismissed on the same day, returning for additional scheduled encounters.

Prospective pilot study — A preliminary study investigating the feasibility of an intervention, usually preceding the conduction of a Randomized Controlled Trial.

Randomized Controlled Trial — A type of design of clinical trials that follows controlled guidelines to prevent several forms of bias, and are double blinded, that is, both participants and research team are blinded to conditions and outcomes tested with a control group, in a between-subjects design.

Randomized Experimental Study — Studies composed with a main pharmacological intervention, randomized but not necessarily with placebo blinding control.

Self-titrated condition — A condition in which participants take their medication (or placebo) as they wish, usually with a determined maximum dose (i.e., puffs from a spray, maximum of 10 puffs a day).

Smoking as Usual condition — A condition in which participants are asked to smoke at their usual rate, usually to establish a baseline condition of smoking before intervention.

Systematic Review — A reproducible method of review of a formulated question, following the 'PICO' format (Population, Intervention, Comparison and Outcomes) to assess the quality of Randomized Controlled Trials and compile high quality data in order to provide a Meta Analysis on a subject.

Within-subjects design — A design of intervention in which the same participant is exposed to all the conditions (i.e., all participants take medication and placebo).

Declaration of Competing Interest

None.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.addicn.2022.100025.

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